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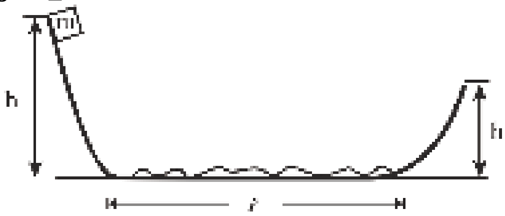
MD

PHYSICS

1) A body of mass m starting from rest from origin moves along x-axis with constant power (P). Then relation between velocity and time is :-

(1) $v \propto t^{1/2}$
(2) $v \propto t^{2}$
(3) $v \propto t^{3}$
(4) $v \propto t^{3/2}$

2) A block is released from rest from a height h = 5m. After travelling through the smooth curved surface it moves on the rough horizontal surface through a length ℓ = 8 m and climbs onto the other smooth curved surface through a height h'. If μ = 0.5, find h' :-

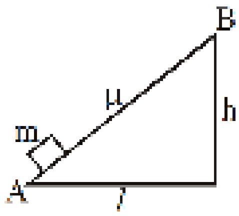


(1) 2 m
(2) 3 m
(3) 1 m
(4) zero

3) A uniform flexible chain of mass m and length 2ℓ hangs in equilibrium over a smooth horizontal pin of negligible diameter. One end of the chain is given a small vertical displacement so that the chain slips over the pin. The speed of chain when it leaves pin is

(1) $\sqrt{2g\ell}$
(2) $\sqrt{g\ell}$
(3) $\sqrt{4g\ell}$
(4) $\sqrt{3g\ell}$

4) Amount of work done to carry a block from A to B will be (Assume friction coefficient μ)



(1) mgh
(2) $\mu mg\sqrt{l^2 + h^2}$

(3) $\mu mg(l + h)$

(4) $mg(h + \mu l)$

5) A body of mass m = 10^{-2} kg is moving in a medium and experience a frictional force F = $-kv^2$. its initial speed is v^0 = 10 ms^{-1}. After 10s, its kinetic energy is $\frac{1}{8}mv_0^2$, then value of k will be :-

(1) 10^{-1} kg m^{-1} s^{-1}

(2) 10^{-3} kg m^{-1}

(3) 10^{-3} kg s^{-1}

(4) 10^{-4} kg m^{-1}

6) The potential energy of a particle of mass 5 kg moving in the x – y plane is given by U = (–7x + 24y)J, where x and y are given in metre. If the particle starts from rest, from the origin, then the speed of the particle at t = 2 s is

(1) 5 m/s

(2) 14 m/s

(3) 17.5 m/s

(4) 10 m/s

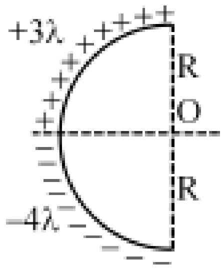
7) The existence of a negative charge on a body implies that it has :-

(1) lost some of its electrons

(2) lost some of its protons

(3) acquired some electrons from outside

(4) acquired some protons from outside



8) Net electric field intensity at centre O is –

(1) $\frac{2k\lambda}{R}$

(2) $\frac{\sqrt{2}k\lambda}{R}$

(3) $\frac{5k\lambda}{R}$

(4) $\frac{5\sqrt{2}k\lambda}{R}$

9) Let A and B are two identical objects of same mass. initially are charged by rubbing against each other. Let A has got positive charge and B has gets negative charge then match column-I and column-II

Column-I		Column-II	
i.	mass of A	a.	increases
ii.	mass of B	b.	decreases
iii.	Work function of A	c.	is less
iv.	work function of B	d.	is more

(1) i-d, ii-c, iii-b, iv-a

(2) i-b, ii-a, iii-d, iv-c

(3) i-b, ii-a, iii-c, iv-d

(4) i-c, ii-d, iii-a, iv-b

10) A small circular ring has a uniform charge distribution. On a far-off axial point distance x from the centre of the ring, the electric field is proportional to :-

(1) x^{-1}

(2) $x^{-3/2}$

(3) x^{-2}

(4) $x^{5/4}$

11) Two point charges –4Q and 9Q are placed at a distance 2m from each other. The position at which net electric field is zero from the charge –4Q is –

(1) 3m

(2) 4m

(3) 2m

(4) 1m

12) Two small spheres have mass m_1 and m_2 and hanging from massless insulating threads of lengths ℓ_1 and ℓ_2. Two sphere carry charges Q_1 and Q_2 respectively. The spheres hang such that they are on the same horizontal level and the threads are inclined to the vertical at angle θ_1 and θ_2 respectively. Which of the following condition is true if $m_1 = m_2$.

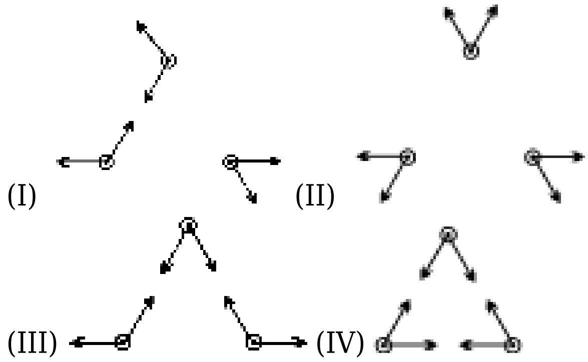
(1) $\theta_1 = \theta_2$

(2) $q_1 = q_2$

(3) $\frac{\ell_1}{\tan\theta_1} = \frac{\ell_2}{\tan\theta_2}$

(4) $\frac{q_1}{\tan\theta_1} = \frac{q_2}{\tan\theta_2}$

13) The figure below shows the forces that three charged particles exert on each other. Which of the four situations shown can be correct.



(1) all of the above

(2) None of the above

(3) II, III

(4) II, III & IV

14) -1×10^{-6} C charge is on a drop of water having mass 10^{-6} kg. What electric field should be applied on the drop so that it is in the balanced condition with its weight ?

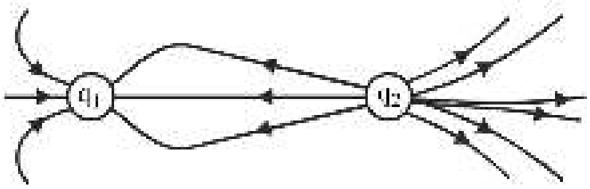
(1) 10 V/m upward

(2) 10 V/m downward

(3) 0.1 V/m downward

(4) 0.1 V/m upward

15) For the figure shown, what is the ratio of the charges $\frac{|q_1|}{|q_2|}$, where the figure shown has a representation of the field lines in the space near the charges.



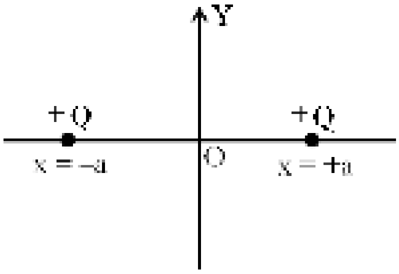
(1) 4/3

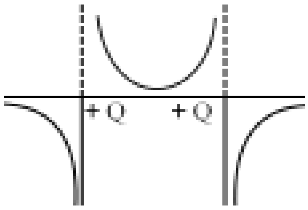
(2) 1/3

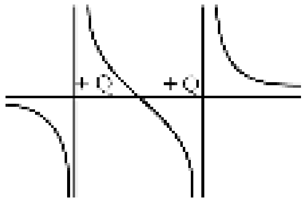
(3) 2/3

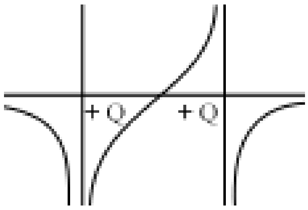
(4) 3/2

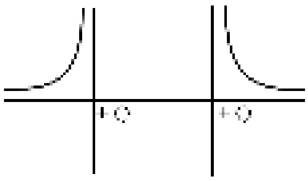
16) Two identical positive point charge placed on x-axis the variation of E along x-axis.



(1) 

(2) 

(3) 

(4) 

17) ***Assertion :*** In electrostatics electric field lines are always perpendicular to surface of charged metallic conductor.
Reason : In electrostatics, surface of charged metallic conductor is always equipotential surface.

(1) If both Assertion & Reason are true & the reason is a correct explanation of the Assertion.
(2) If both Assertion & Reason are true but reason is not a correct explanation of the Assertion.
(3) If Assertion is true but the Reason is false
(4) If both Assertion & Reason are false

18) (a) Electric line of force can not crossed metal surface.
(b) Electric line of force perpendicular to metal surface.
(c) On a metal surface distribution of charge is always uniform.
(d) On spherical metal surface distribution of charge may be nonuniform.
In above statements which of following will be correct :-

(1) a and b
(2) a, b, c
(3) a, b, d
(4) all a, b, c, d

19) Two small identical spheres having charges +10μC and –90μC attact each other with a force of F newton. If they are kept in contact and then separated by the same distance, the new force between them is :-

(1) $\frac{F}{6}$
(2) 16 F

(3) $\frac{16F}{9}$

(4) 9F

20) 

In given system of point charge which of following statement is correct.
(a) For any value of q_0, q_0 charge will be in equilibrium.
(b) For any value of q_0 system will be in equilibrium.
(c) If q_0 is in equilibrium then q_1 and q_2 are like.
(d) If q_1 and q_2 are unlike then given system can not be in equilibrium.

(1) a and b

(2) a and c

(3) a, c, d

(4) a, d

21) 

For what value of q given system will be in equilibrium.

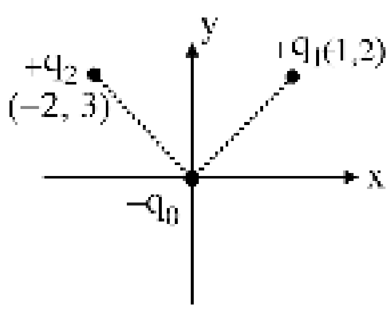
(1) $\frac{9Q}{16}$

(2) $\frac{-9Q}{4}$

(3) $\frac{-9Q}{16}$

(4) $\frac{-Q}{4}$

22) In given system of point charges, $+q_1$ is placed at (1, 2) and $+q_2$ is placed at (–2, 3). If net force



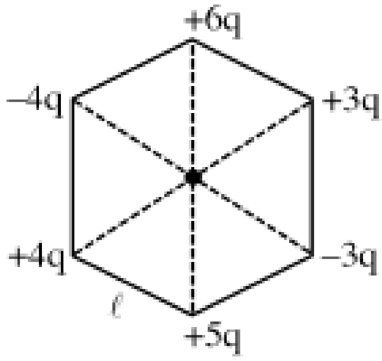
on $-q_0$ is along + y then $\frac{q_1}{q_2}$ will be

(1) $\frac{10}{13}\sqrt{\frac{5}{13}}$

(2) $10\frac{\sqrt{5}}{13}$

(3) $\frac{\sqrt{5}}{13}$

(4) $\frac{13}{5}$



23) The magnitude of electric field at the centre of regular hexagon.

(1) $\frac{kq}{\ell^2}$

(2) $\frac{\sqrt{2}Kq}{\ell^2}$

(3) $2+\sqrt{2}$

(4) zero

24) From the photo electric effect experiment following observations are made, identify which of these are correct :
(A) The stopping potential depends only on the work function of the metal.
(B) The saturation current increases as the intensity of incident light of given frequency increases.
(C) The maximum kinetic energy of a photoelectron depends on the intensity of the incident light.
(D) Photoelectric effect can be explained using wave theory of light.
Choose the correct answer from the options given below :

(1) B, C only

(2) A, C, D only

(3) B only

(4) A, B, D only

25) A nucleus with mass number 242 and binding energy per nucleon as 7.6 MeV breaks into fragments each with mass number 121. If each fragment nucleus has binding energy per nucleon as 8.1 MeV the total gain in binding energy is _____ MeV

(1) 121

(2) 250

(3) 50

(4) 43

26) The de-Broglie wavelength of a molecule in a gas at room temperature (300 K) is λ_1. If the temperature of the gas is increased to 600 K then the de-Broglie wavelength of the same gas molecule becomes :

(1) $\frac{1}{\sqrt{2}}\lambda_1$

(2) $2\lambda_1$

(3) $\frac{1}{2}\lambda_1$

(4) $\sqrt{2}\lambda_1$

27) A nucleus disintegrates into two smaller parts, which have their velocities in the ratio 3 : 2. The ratio of their nuclear sizes will be $\left(\frac{x}{3}\right)^{\frac{1}{3}}$. The value of x is :

(1) 1
(2) 2
(3) 3
(4) 4

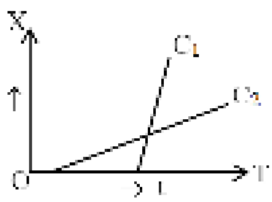
28) A radioactive element $^{242}_{92}X$ emits two α-particles one electron and two positrons. The product nucleus is represented by $^{234}_{P}Y$. The value of P is :

(1) 85
(2) 84
(3) 86
(4) 87

29) Monochromatic light of frequency 6×10^{14} Hz is produced by a laser. The emitted power is 2×10^{-3} W. How many photons per second on an average are emitted by the source [$h = 6.63 \times 10^{-34}$]

(1) 9×10^{18}
(2) 6×10^{15}
(3) 5×10^{15}
(4) 7×10^{16}

30) Shown in the figure are the position time graph for two children going home from the school. Which of the following statements about their relative motion is true after both of them started moving?



There relative velocity : (consider 1-D motion)

(1) first increases and then decreases
(2) first decreases and then increases
(3) is zero
(4) is non zero constant

31) A body is thrown up in a lift with a velocity u relative to the lift and the time of flight is found to be 't'. The acceleration with which the lift is moving up will be :

(1) $\frac{u - gt}{t}$

(2) $\frac{u+gt}{t}$

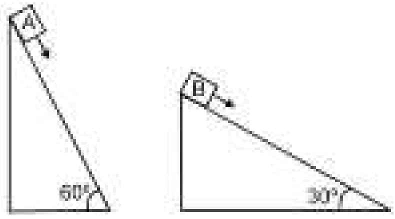
(3) $\frac{2u-gt}{t}$

(4) $\frac{2u+gt}{t}$

32) For four particles A, B, C & D, the velocities of one with respect to other are given as $\vec{V}_{DC}$ is 20 m/s towards north, $\vec{V}_{BC}$ is 20 m/s towards east and $\vec{V}_{BA}$ is 20 m/s towards south. Then $\vec{V}_{DA}$ is :

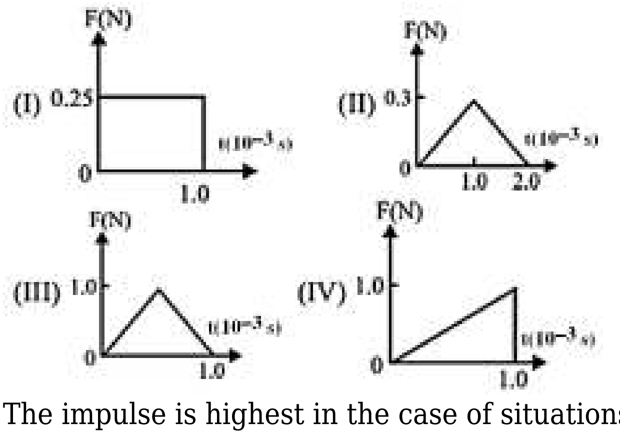
(1) 20 m/s towards north
(2) 20 m/s towards south
(3) 20 m/s towards east
(4) 20 m/s towards west

33) Two fixed frictionless inclined planes making an angle 30° and 60° with the vertical are shown in the figure. Two blocks A and B are placed on the two planes. What is the relative vertical acceleration of A with respect to B ?



(1) 4.9 ms^{-2} in horizontal direction
(2) 9.8 ms^{-2} in vertical direction
(3) zero
(4) 4.9 ms^{-2} in vertical direction

34) Figure I, II, III and IV depict variation of force with time :



The impulse is highest in the case of situations depicted. Figure :

(1) I and II
(2) III and I
(3) III and IV
(4) IV only

35) Which one of the following statements is not true ?

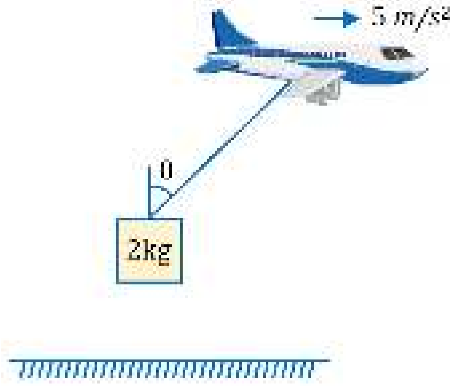
(1) The same force for the same time causes the same change in momentum for different bodies.

(2) The rate of change of momentum of a body is directly proportional to the applied force and takes place in the direction in which the force acts.

(3) A greater opposing force is needed to stop a heavy body than a light body in the same time, if they are moving with the same speed.

(4) The greater the change in the momentum in a given time, the lesser is the force that needs to be applied.

36) An airplane is moving horizontally with constant acceleration 5 m/s^2 with a block of mass $2kg$ is hanging with it, which is at rest w.r.t. plane find angle θ which string makes with vertical?



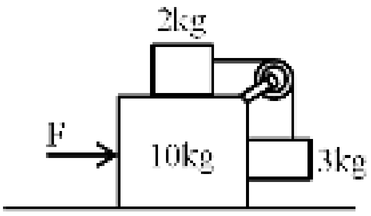
(1) $\tan\theta = 2$

(2) $\cot\theta = 2$

(3) $\sin\theta = \frac{1}{2}$

(4) $\cos\theta = \frac{1}{2}$

37) All surfaces are smooth. Find value of F so that there is no relative motion between 2 kg and 10 kg.

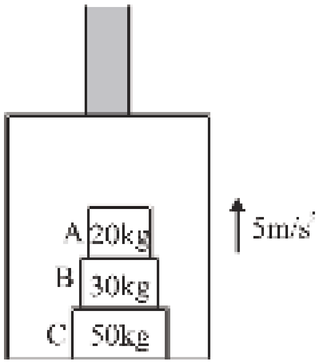


(1) 200 N

(2) 225 N

(3) 125 N

(4) None of these

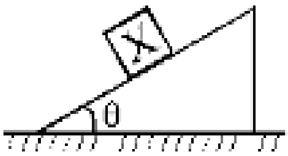


38) In shown situation elevator is moving upward with acceleration of 5 m/s^2.

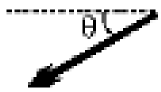
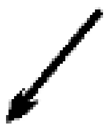
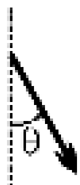
	Column I		Column II
(1)	Net force acting on B	(P)	150 N
(2)	Normal reaction between A and B	(Q)	300 N
(3)	Normal reaction between B and C	(R)	450 N
(4)	Normal reaction between C and elevator	(S)	750 N
		(T)	1500 N

(1) (1)-P (2)-Q (3)-S (4)-T
(2) (1)-Q (2)-Q (3)-R (4)-T
(3) (1)-P (2)-R (3)-S (4)-S
(4) (1)-Q (2)-R (3)-R (4)-T

39) In the situation shown, all surfaces are frictionless and triangular wedge is free to move. In list-II the direction of certain vectors are shown. Match the direction of quantities in list-I with possible



vector in list-II.

List-I		List-II	
(A)	Acceleration of the block X relative to ground	(P)	
(B)	Acceleration of block X relative to wedge	(Q)	
(C)	Normal force by block on wedge	(R)	
(D)	Net force on the wedge	(S)	

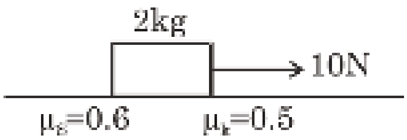
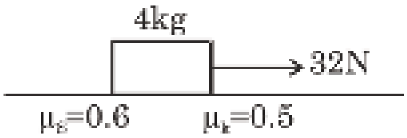
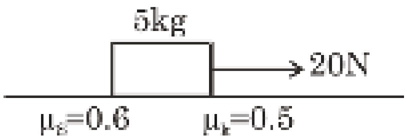
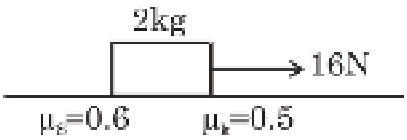
(1) A→Q; B→P; C→R; D→S

(2) A→S; B→P; C→Q; D→R

(3) A→P; B→Q; C→S; D→R

(4) A→R; B→S; C→P; D→Q

40) In **List-I** blocks are initially at rest on rough horizontal floor, and forces are applied on the blocks at t = 0 as shown. **List-II** represents friction, acceleration and speed of blocks. Match the **correct** option(s) of both list : (g = 10 m/s^2)

List-I		**List-II**	
(P)		(1)	Friction acting on block is 20 N
(Q)		(2)	Acceleration of block is zero
(R)		(3)	Speed of block after 3 sec is 9 m/s
(S)		(4)	Friction acting on block is 10 N
		(5)	Acceleration of block is 3 m/s^2

(1) P → 2,4; Q → 1,3,5; R → 1,2; S→ 3,4,5

(2) P → 2,4; Q → 2,3,5; R → 1,3; S → 1,2,3

(3) P → 1,5; Q → 1,3,5; R → 2,4; S → 2,4

(4) P → 3,4; Q → 3,4,5; R → 4,5; S → 2,3,5

41) **Assertion (A) :-** Magnitude of contact force is always greater than the magnitude of non-zero friction force.
Reason (R) :- Contact force is the resultant of the friction force & normal reaction.

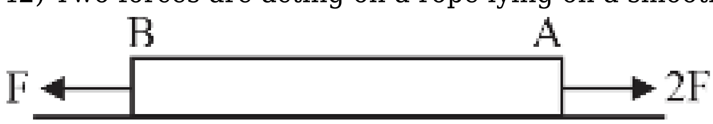
(1) Both Assertion & Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion & Reason are true but Reason is not correct explanation of Assertion.

(3) Assertion is true but Reason is false

(4) Both Assertion & Reason are false

42) Two forces are acting on a rope lying on a smooth table as shown in figure.



Which of the following statement(s) is/are correct?

I. In moving from A to B, tension on string decreases from 2F to F.
II. Situation will becomes indeterminant, if we take it a massless string.

(1) Only I
(2) Only II
(3) Both I and II
(4) None of these

43)

On a planet particle is thrown with a velocity $\vec{V} = 6\hat{i} + (20 - 4t)\hat{j}$. Then match the list.

List-I		List-II	
(P)	Time of flight	(1)	3
(Q)	Time when particle is moving at an angle of 53° with horizontal in upwards direction.	(2)	10
(R)	Range of the particle	(3)	50
(S)	Maximum height of the particle	(4)	60

(1) P → 2; Q → 1; R → 4; S → 3
(2) P → 1; Q → 2; R → 3; S → 4
(3) P → 2; Q → 3; R → 4; S → 1
(4) P → 2; Q → 4; R → 1; S → 3

44) A particle is projected obliquley from horizontal surface it just clear a goal post as shown in

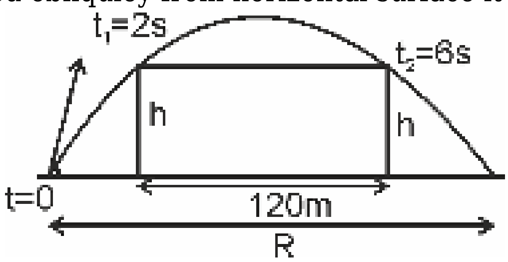
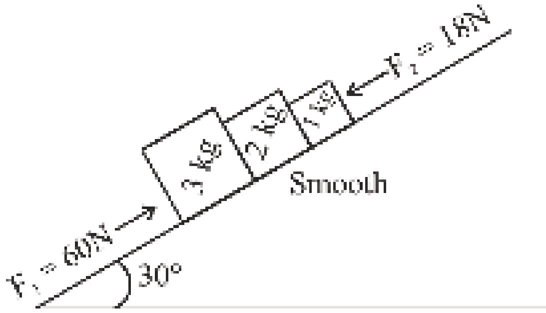


figure then its range is :

(1) 240 m
(2) 420 m
(3) 120 M
(4) None of these

45) In the diagram shown, match the following (g = 10 m/s^2) Blocks are on smooth incline. F_1 and F_2 are parallel to the inclined plane. The motion of the blocks is along the incline the surface.



Column-I		Column-II	
(a)	Acceleration of 2 kg block	(p)	39 SI unit
(b)	Net force on 3 kg block	(q)	25 SI unit
(c)	Normal reaction between 2 kg and 1 kg	(r)	2 SI unit
(d)	Normal reaction between 3 kg and 2 kg	(s)	6 SI unit

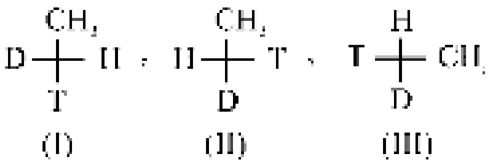
(1) a → r; b → p; c → q; d → s

(2) a → p; b → r; c → q; d → s

(3) a → q; b → p; c → r; d → s

(4) a → s; b → p; c → q; d → r

CHEMISTRY

1) 

Choose the correct statement

(1) I & II have R configuration

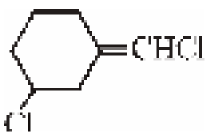
(2) I & III are enantiomers

(3) II & III are enantiomers

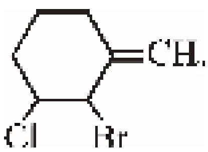
(4) all are identical

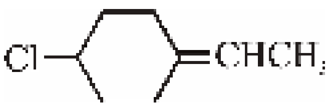
2)

Which of the following does not show geometrical isomerism ?

(1) 

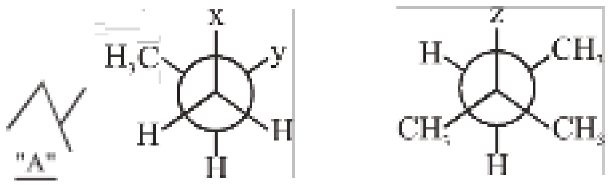
(2) 

(3) 

(4) 

3)

Consider following molecule “A” & find x, y & z in given newman’s projection respectively:



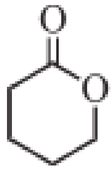
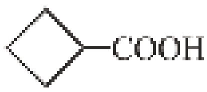
(1) CH_3, CH_3 and H

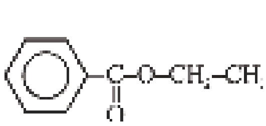
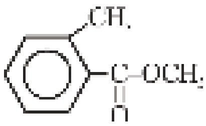
(2) CH_3, H and CH_3

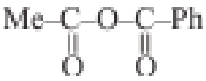
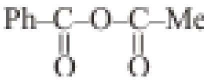
(3) H, CH_3 and CH_3

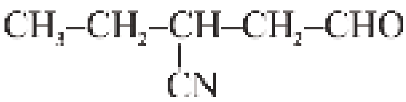
(4) CH_3, CH_3 and CH_3

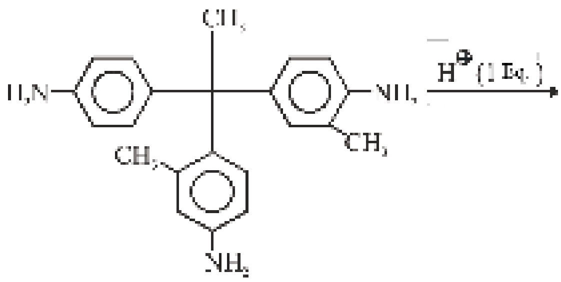
4) Which of the following statement is NOT correct:-

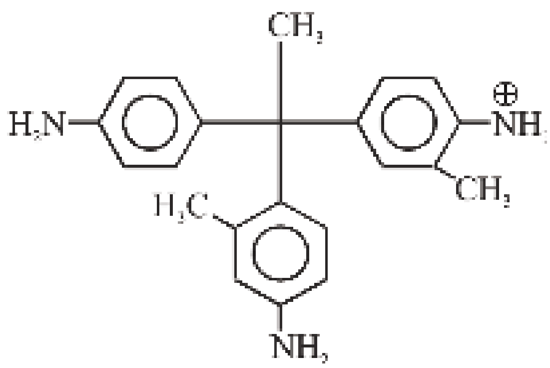
(1)  and 
are functional group isomers

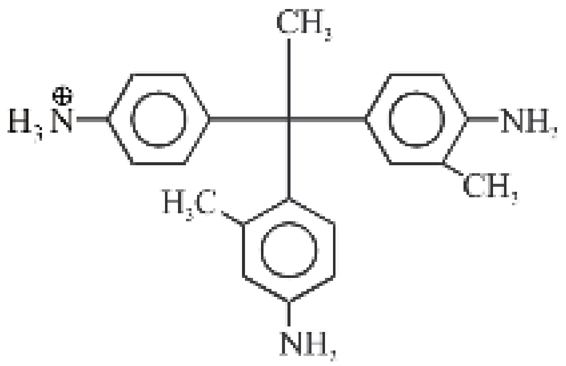
(2)  and 
are metamers

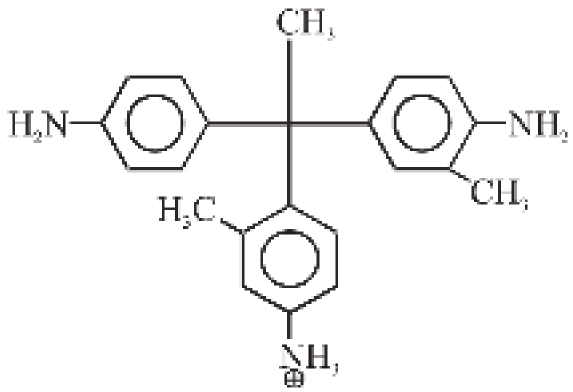
(3)  and 
are metamers

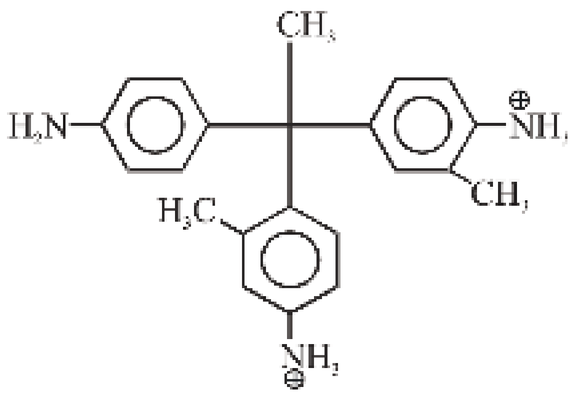
(4)  and 
are chain isomers

5) 

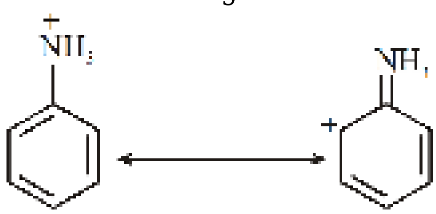
(1) 

(2) 

(3) 

(4) 

6) Consider the following two structure of anilinium ion and select the correct statement about



(I) (II)

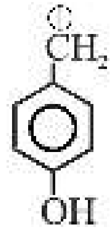
them:-

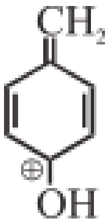
(1) II is not an acceptable canonical structure because carbocation are less stable than ammonium ions.

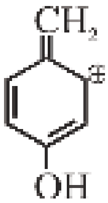
(2) II is not an acceptable canonical structure because it is non aromatic.

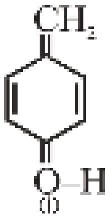
(3) II is not an acceptable canonical structure because the nitrogen has 10 valence electrons.

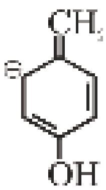
(4) II is an acceptable canonical structure.

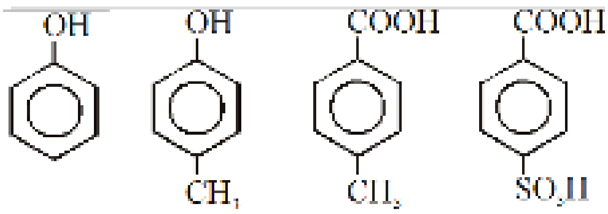
7) Which is most stable resonating structure of  :-

(1) 

(2) 

(3) 

(4) 



(a) (b) (c) (d)

8)
Compare acidic strength :-

(1) a > c > d > b

(2) d > a > c > b

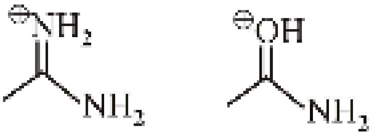
(3) d > c > a > b

(4) d > c > b > a

9) Which cationic species is more stable in each of the following pairs ?

Pair X 

1 2

Pair Y 

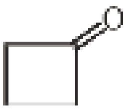
3 4

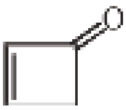
(1) 1 and 3
(2) 1 and 4
(3) 2 and 4
(4) 2 and 3

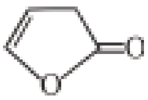
10) Which of the following is correct decreasing order of Acidic strength :-

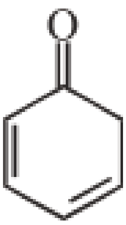
(1) $HCOOH > CH_3COOH > H_2CO_3 > C_6H_5COOH$
(2) $HCOOH > C_6H_5COOH > CH_3COOH > H_2CO_3$
(3) $HCOOH > H_2CO_3 > CH_3COOH > C_6H_5COOH$
(4) $CH_3COOH > HCOOH > C_6H_5COOH > H_2CO_3$

11) Which of following compounds forms highly unstable enol ?

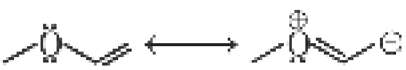
(1) 

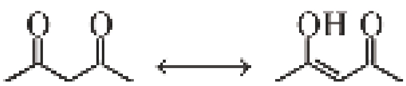
(2) 

(3) 

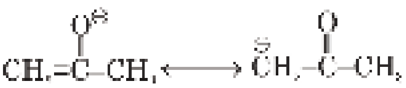
(4) 

12) Which are not the pair of resonating structure ?

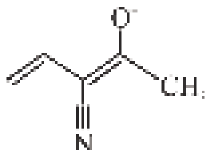
(1) 

(2) 

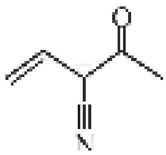
(3) $N\equiv C-O^{\ominus} \longleftrightarrow \overset{\ominus}{N}=C=O$

(4) 

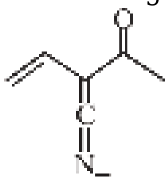
13) Compare relative stability of the following resonating structure :



(p)



(q)



(r)

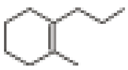
(1) p > q > r

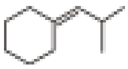
(2) q > p > r

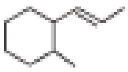
(3) q > r > p

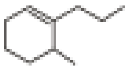
(4) p > r > q

14) Which of the following shows minimum heat of combustion ?

(1) 

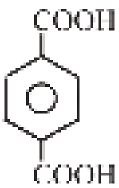
(2) 

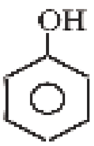
(3) 

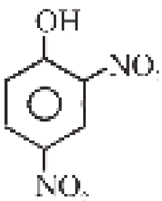
(4) 

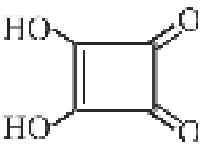
15)

Which of the following not give CO_2 gas with $NaHCO_3$?

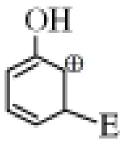
(1) 

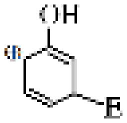
(2) 

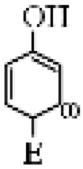
(3) 

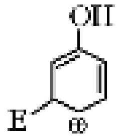
(4) 

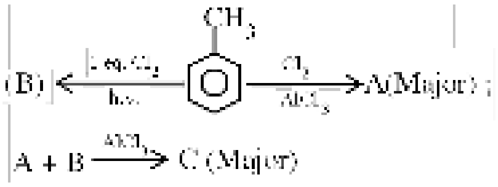
16) Which σ complex is most stable :

(1) 

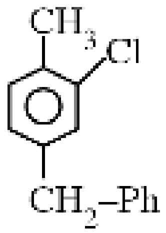
(2) 

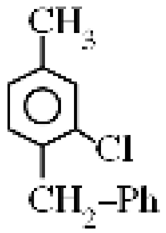
(3) 

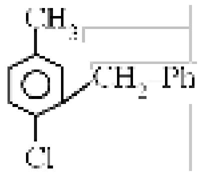
(4) 

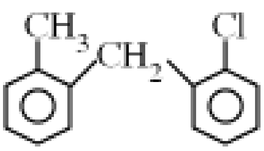
17) 

Major product "C" in above reaction sequence:

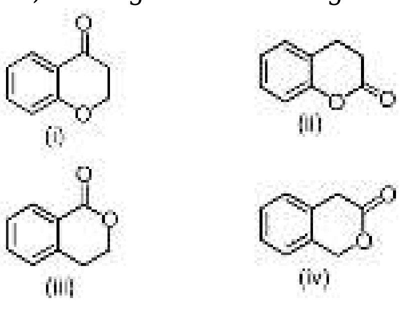
(1) 

(2) 

(3) 

(4) 

18) Arrange the following in decreasing order of reactivity towards bromination



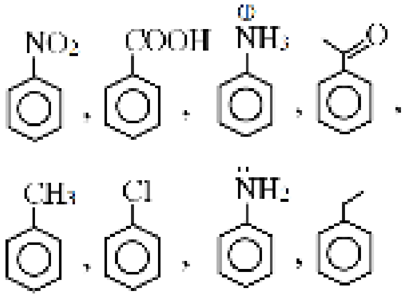
(1) ii > i > iv > iii

(2) iv > ii > i > iii

(3) i > ii > iii > iv

(4) iv > iii > ii > i

19) How many compounds Not show Friedel-Craft reaction :



(1) 3

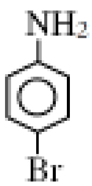
(2) 4

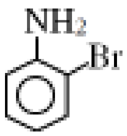
(3) 5

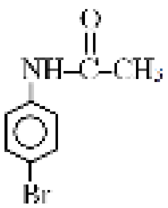
(4) 6

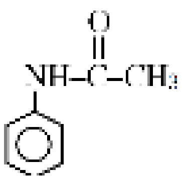
20) 

Find out C :

(1) 

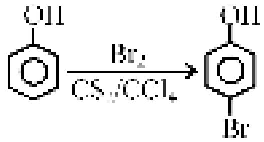
(2) 

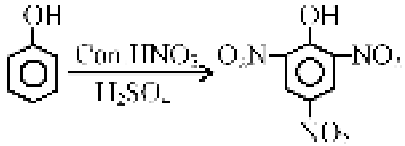
(3) 

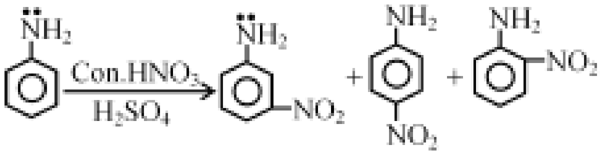
(4) 

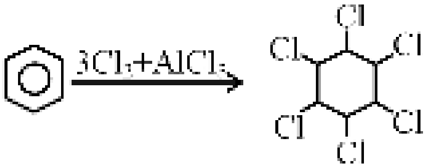
21)

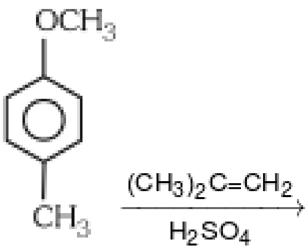
Which reaction show incorrect product.

(1) 

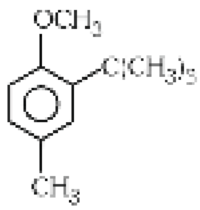
(2) 

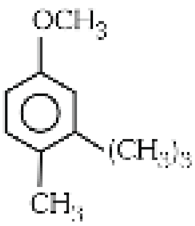
(3) 

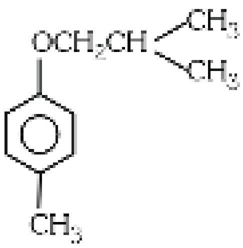
(4) 

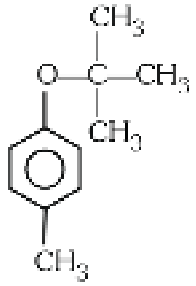
22)  Product

product is :-

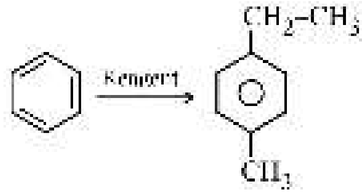
(1) 

(2) 

(3) 

(4) 

23) Give the correct order of reagent in following reaction respectively :-



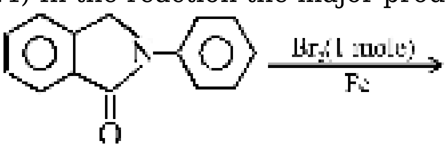
(1) $CH_3-\overset{O}{\overset{\|}{C}}-Cl + AlCl_3$, CH_3 – Cl + $AlCl_3$,
Zn – Hg + HCl

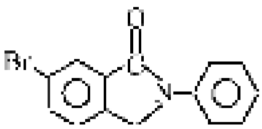
(2) Zn – Hg / HCl, CH_3 – Cl + $AlCl_3$,
$CH_3-\overset{O}{\overset{\|}{C}}-Cl + AlCl_3$

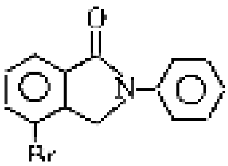
(3) $CH_3-\underset{O}{\underset{\|}{C}}-Cl-AlCl_3$, Zn-Hg/HCl, CH_3 – Cl + $AlCl_3$

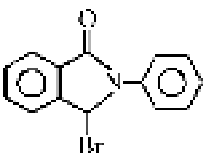
(4) All are correct

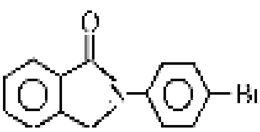
24) In the reaction the major product formed is

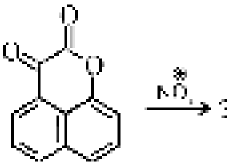


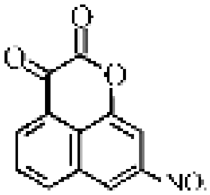
(1) 

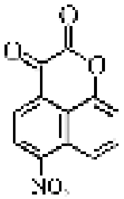
(2) 

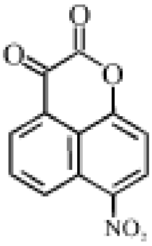
(3) 

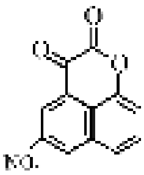
(4) 

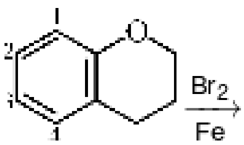
25)  :-

(1) 

(2) 

(3) 

(4) 

26)  Substitution takes place at the position.

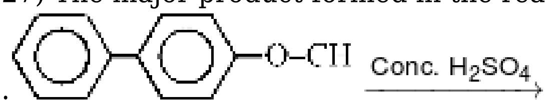
(1) 1

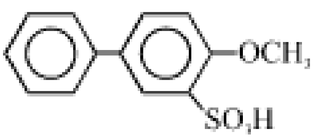
(2) 2

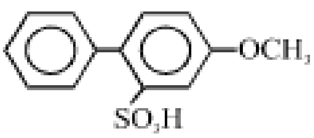
(3) 3

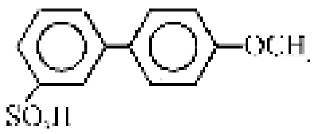
(4) Both (1) and (3)

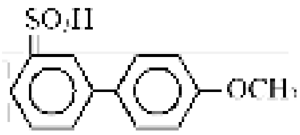
27) The major product formed in the reaction



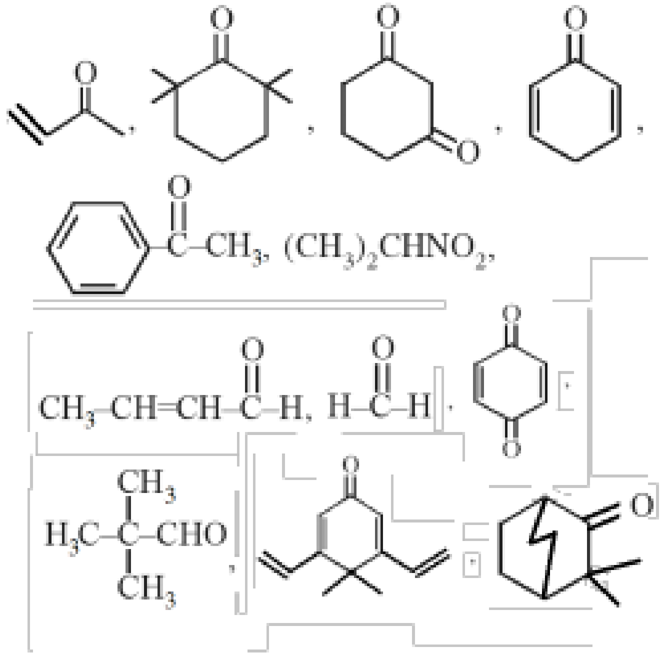
(1) 

(2) 

(3) 

(4) 

28) Number of compound which show tautomerism :-



(1) 7

(2) 6

(3) 5

(4) 9

29) Number of moles of Mn_3O_4 that will require to react completely with 1 mol of $FeC_2O_4.Fe_2(C_2O_4)_3$ in presence of H_2SO_4

(1) 9 mol

(2) 9/5 mol

(3) 4.5 mol

(4) $\frac{11}{2}$ mol

30) What will be the EMF for the given cell ?
$Pt(|H_2(g),P_1)|H^+(aq,1M)||H^+(aq.,1M)|(H_2(g),P_2)|Pt(s)$

(1) $\frac{RT}{F} \ln \frac{P_1}{P_2}$

(2) $\frac{RT}{2F} \ln \frac{P_1}{P_2}$

(3) $\frac{RT}{F} \ln \frac{P_2}{P_1}$

(4) $\frac{RT}{2F} \ln \frac{P_2}{P_1}$

31) The standard reduction potential values of three metallic cations, X,Y,Z are 0.52, –3.03 and –1.18 V respectively. The order of reducing power of the corresponding metals is :-

(1) Y > Z > X
(2) X > Y > Z
(3) Z > Y > X
(4) Z > X > Y

32) When an aqueous solution of $CuSO_4$ is stirred with a silver spoon then :-

(1) Cu^+ will be formed
(2) Ag^+ will be formed
(3) Cu^{2+} will be deposited
(4) None of these

33) What will be the theoretical change in cell potential of a cell $Ni|Ni^{2+}_{(0.001M)}||Ag^{1+}_{(0.01M)}|Ag$ when its temperature is increased

(1) increase
(2) decrease
(3) no change
(4) cannot be said

34) Given that $E^o_{Zn^{+2}|Zn}$ = –0.763 volt and $E^o_{Fe^{+2}|Fe}$ = –0.44 volt, then the E.M.F. of the cell $Zn_{(s)}\left|Zn^{+2}\right.$ (0.001 M) $\left\|Fe^{+2}\,(0.005M)\right|Fe_{(s)}$ is :-

(1) Equal to 0.323 volt
(2) Less than 0.323 volt
(3) Greater than 0.323 volt
(4) None of these

35) How much will the oxidation potential of a hydrogen electrode change when its solution initially at pH = 0 is neutralised to pH = 7 ?

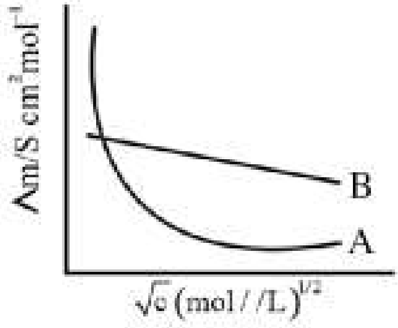
(1) increases by 0.059 V
(2) decreases by 0.059 V
(3) increases by 0.413 V
(4) decreases by 0.413 V

36) Calculate $E^\circ_{MnO_2/MnO_4^-}$ given

$E^{\circ}_{Mn^{+2}/MnO_4^-}$ = –1.51V and
$E^{\circ}_{MnO_2/Mn^{+2}}$ = 1.23V

(1) –1.69V
(2) 1.69
(3) 0.28
(4) –0.28

37) Following figure shows dependence of molar conductance of two electrolytes on concentration.



Λ_m^0 is the limiting molar conductivity. The number of Incorrect statement(s) from the following is ________

(1) Λ_m^0 for electrolyte A is obtained by extrapolation
(2) For electrolyte B, Λm vs $\sqrt{c}$ graph is a straight line with intercept equal to Λ_m^0
(3) At infinite dilution, the value of degree of dissociation approaches maximum for electrolyte B.
(4) Λ_m^0 for any electrolyte A or B can be calculated using λ° for individual ions.

38) **Assertion:** In Daniell cell, electrons flow from Zn rod to Cu rod.
Reason: Zn rod acts as cathode and Cu rod acts as anode in Daniell cell.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

39) Saturated solution of KNO_3 is used to make 'salt-bridge' because :

(1) velocity of K^+ is greater than that of NO_3^-
(2) velocity of NO_3^- is greater than that of K^+
(3) velocity of both K^+ and NO_3^- are nearly the same
(4) KNO_3 is high soluble in water

40) **Assertion :** Specific conductivity of all electrolytes decreases on dilution.
Reason : On dilution number of ions per unit volume decreases.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion
(2) Both Assertion and Reason are true but reason is not correct explanation of Assertion

(3) Assertion is true but Reason is false

(4) Assertion is false but Reason is true

41) Given $\Lambda_{eq}^{\infty} BaCl_2 = x\, Scm^2 eq^{-1}$

$\Lambda_{eq}^{\infty} H_2SO_4 = y\, S\,cm^2 eq^{-1}$

$\Lambda_{m}^{\infty} HCl = z\, S\,cm^2 eq^{-1}$

correct value of Λ_{eq}^{∞} $BaSO_4$ is

(1) x+2y–z Scm^2eq^{-1}

(2) (x+y–z) Scm^2eq^{-1}

(3) x+y–2z Scm^2eq^{-1}

(4) 2x+y–z Scm^2eq^{-1}

42) The metal that can not be obtained by electrolysis of an aqueous solution of its salts is :

(1) Zn

(2) Mn

(3) Cd

(4) Co

43) For the cell reaction with the indicated concentrations

$Al_{(s)} + 3Ag^{+}_{(aq)}(0.10\,M) \rightarrow Al^{3+}_{(aq)}(0.30M) + 3Ag_{(s)}$, the measured voltage ($E_{cell}$) is 1.50 volt. Calculate E°_{cell}.

Given : $\dfrac{2.303\,RT}{F}$ = 0.06 ; log 3 = 0.48

(1) 1.5496 volt

(2) 1.654 volt

(3) 1.4032 volt

(4) None of these

44) The emf of the cell,

$Tl_{(s)}|Tl^{+}_{(aq)}$ (0.0001 M)|| $Cu^{2+}_{(aq)}$(0.01M) | $Cu_{(s)}$ is 0.83 V

The emf of this cell will be increased by :–

(1) Increasing the concentration of Cu^{++} ions

(2) Decreasing the concentration of Tl^{+}

(3) Increasing the concentration of both

(4) (1) & (2) both

45) From the following data -

(A) For same conductivity cell, cell constant (G^{*}) will not change.

(B) λ_m = K × V; where V is volume of solution in which 1g equivalent electrolytic solute is present; K is specific conductivity & λ_m is molar conductivity.

(C) Ionic mobility of $H^{+}_{(aq)}$ is less than $Li^{+}_{(aq)}$

(D) Kohlrousch law is applicable for all type of electrolytes at infinite dilution.
(E) E°_{cell} depends on stoicheometry coefficient of cell reaction.
(F) In conductivity cell AC source is used because in DC; deposition of ions may occur.
(G) At 600K; E_{cell} can be calculated by Nernst equation.

$$E_{cell} = E^0_{cell} - \frac{0.06}{n} \log \frac{[RHS]}{[LHS]}$$

The correct statements is :

(1) 3
(2) 5
(3) 6
(4) 2

BIOLOGY

1) Match the following three columns giving some information/detail about the vertebrate's heart.

	Column I		Column II		Column III
i	Pisces	a.	3 chambered	A	One atrium, & Two ventricles
ii	Amphibians	b.	2 chambered	B	2 atria, & 1 ventricle
iii	Reptiles	c.	4 chambered	C	1 atrium, & 1 ventricle
iv	Aves	d.	single chambered	D	2 atria, & 2 ventricles

(1) i-b-C, ii-c-D, iii-a-B, iv-c-B
(2) i-b-B, ii-a-B, iii-a-C, iv-c-D
(3) i-b-B, ii-b-C, iii-b-C, iv-a-B
(4) i-b-C, ii-a-B, iii-a-B, iv-c-D

2) Choose the correct and wrong statements about the artery :-
a. carry deoxygenated blood without any exception
b. carry blood towards heart
c. carry oxygenated blood with exception
d. carry blood from heart towards organs
e. is highly muscular

(1) Correct a, b, c wrong d, e
(2) Correct c, d, e wrong a, b
(3) Correct a, c, e wrong d, b
(4) Correct a, b, d wrong c, e

3)

Read the following statement :-
(A) The hepatic portal vein carries blood from intestine to the liver.
(B) Adrenal medullary hormones can also decrease the cardiac output.

(C) During a cardiac cycle, both ventricle pumps out approximately 5 L of blood.
How many statements are correct ?

(1) One
(2) Two
(3) Three
(4) All

4)

Choose the correct option for X, Y, Z and P :-

Blood Group	May receive blood from	May donate blood to
O	O	Z
A	X	A, AB
B	B, O	B, AB
AB	Y	P

	X	Y	Z	P
(1)	A,O	A,B,AB,O	A,B,AB,O	AB
(2)	A,O	A,B,AB,O	A,B,AB,O	A,B
(3)	B	A,B,AB,O	A,B,AB,O	AB
(4)	AB	A,B, AB,O	A,B,AB,O	AB

(1) 1
(2) 2
(3) 3
(4) 4

5) Read the following properties of blood corpuscles:-
(a) They are colourless due to lack of haemoglobin
(b) They are non-nucleated
(c) They are relatively lesser in number which averages 6000-8000 mm^{-3} of blood
(d) They have long life span approx 120 days.
Out of these which properties are not shown by leukocytes ?
Options :-

(1) 'a' and 'b'
(2) 'b' and 'c'
(3) 'b' and 'd'
(4) 'c' and 'd'

6) Which of the following substances are required for the start of coagulation of blood in mammals?

(1) Thrombin and sodium ions
(2) Fibrin and iron ions
(3) Thromboplastin and calcium ions
(4) Heparin and fibrinogen

7) Which of the following statement is wrong ?

(1) Cardiac output of a person varies according to his physiological demand.
(2) Stroke volume of heart is almost equal to cardiac output
(3) Stroke volume is amount of blood pumped by the left ventricle in the systemic aorta in one beat.
(4) End systolic volume is about 50 ml.

8)

Which statement is correct ?
(a) Cardiac output of an athelete is higher than an ordinary man.
(b) Body has the ability to alter the stroke volume as well as the heart rate.
(c) Stoke volume multiplied by the heart rate gives the cardiac output.

(1) a and b
(2) a, b and c
(3) b and c
(4) a and c

9) Correct sequence of events during ventricular systole is ?

(1) Closure of Atrioventricular valves → opening of semilunar vlaves.
(2) Opening of semilunar vlaves → closure of atrioventricular valves.
(3) Closure of semilunar vlaves → opening of valves.
(4) Opening of atrioventricular vlaves → closure of semilunar valves.

10) Match column-I with column-II and select the correct option from below :-

Column I		**Column II**	
(a)	Eosinophil	(i)	Coagulation
(b)	RBC	(ii)	Universal recipient
(c)	AB group	(iii)	Resist infection
(d)	Platelet	(iv)	Contraction of heart
(e)	Systole	(v)	Gas transport

(1) b-v, d-iii, e-iv
(2) c-ii, d-i, e-iv
(3) a-iii, b-v, c-i
(4) a-v, c-i, e-iv

11) Which of the following is correct about 'ECG':-

(1) ECG is a graphical representation of electro-chemical activity of brain.

(2) For the detailed evaluation of heart's function, multiple leads are attached to abdominal region.

(3) By counting the number of QRS complexes that occur in a given time period we can find out heart rate.

(4) ECG obtained from different individuals have different shape for a given lead configuration.

12)

Read the following statements (A-D) :-
(A) Arteries always carry blood away from heart.
(B) Valves are absent in the arteries.
(C) Artery always carry deoxygenated blood.
(D) Lumen of artery is wide.
How many statements are wrong ?

(1) Three

(2) Four

(3) One

(4) Two

13) Tissue fluid (lymph) has :-

(1) same protein content as that in plasma.

(2) same formed element composition as that in Blood.

(3) same mineral distribution as that in plasma.

(4) same large water soluble substances composition as plasma.

14)

I. Proteins contributes 6 - 8% of the blood plasma.
II. Plasma contains very high amount of platelets
III. Plasma without the clotting factors is called serum.
IV. Glucose, amino acids, lipids, etc also present in the plasma.

In the above statements :-

(1) All are correct

(2) Only II is false

(3) Only I is correct

(4) All are false

15) Match the following columns and choose the correct answer from the options given below

Column (I)		**Column (II)**	
(A)	Erythrocytes	(i)	Most abundant white blood cells and the main phagocytic cell of the blood

(B)	Eosinophils	(ii)	Least abundant white blood cells
(C)	Neutrophils	(iii)	Resist infections and are associated with allergic reactions
(D)	Basophils	(iv)	Blood cells that contain hemoglobin and transport oxygen
(E)	Lymphocytes	(v)	Specialized antibody producing white blood cells

	A	B	C	D	E
(1)	iv	iii	v	i	ii
(2)	i	ii	iii	iv	v
(3)	ii	iii	i	v	iv
(4)	iv	iii	i	ii	v

(1) 1
(2) 3
(3) 3
(4) 4

16) Carotid artery carries :-

(1) Impure blood to kideneys
(2) *Oxygenated* blood to *brain*
(3) Impure blood from brain
(4) Oxygenated blood to heart

17) SA node is called pace maker of the heart, because-

(1) It can change contractile activity generated by AV node
(2) It delays the transmission of impulse between the atria and ventricles.
(3) It gets stimulated when it receives neural signal
(4) It initiates and maintains the rhythmic contractile activity of heart.

18) During cardiac cycle about _____ % of ventricular filling occurs prior to atrial contraction ___% ventricular filling occurs due to atrial contraction.

(1) 50, 50
(2) 70, 30
(3) 30, 70
(4) 10, 90

19) The rate of heart beat can be increased by

(1) Sympathetic signal

(2) Adrenal medullary hormone

(3) Para sympathetic signal

(4) Both 1 and 2

20) Time period of cardiac cycle in human heart is 0.8 sec. In which of the following condition this time period is decreased ?
(a) Exercies (b) Infant
(c) Relax condition (d) During food intake

(1) a, b

(2) a, b, c

(3) b, c

(4) d, c

21) During a cardiac cycle, blood pumped by the each ventricle in body is–

(1) 140mL

(2) 70mL

(3) 120mL

(4) 500mL

22) Which of the following is correct representation of hepatic portal system ?

(1) Liver → Hepatic Vein → Inferior vena cava

(2) Intestine → Hepatic portal vein → Liver

(3) Liver → Hepatic portal vein → Intestine

(4) Intestine → Hepatic artery → Liver

23) **Assertion** :- Saline water is not given to patients of hypertension. **Reason** :- Saline water can cause vomiting and may drop blood pressure suddenly causing cardiac arrest.

(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) If Assertion is True but the Reason is False.

(4) If both Assertion & Reason are False.

24)

Open circulatory system is present in
(A) Arthropods (insects)
(B) Annelids
(C) Chordates

(D) Molluscs

(1) C only

(2) C and B

(3) A and B

(4) A and D

25) People who have migrated from the planes to an area adjoining Rohtang pass about six months back

(1) Suffer from altitude sickness with symptoms like nausea, fatigue etc.

(2) Have the usual RBCs count but their haemoglobin has very high binding affinity to O_2.

(3) Have more RBCs and their haemoglobin has a lower binding affinity to O_2.

(4) Are not physically fit to play games like football

26)

Read the given statements.
(A) Breathing volumes are estimated by spirometer
(B) High H^+ in tissue favours O_2 dissociation
(C) Diffusion membrane is made up of three layers
(D) Maximum amount of carbonic anhydrase is found in plasma
Choose the option which includes only the correct statements.

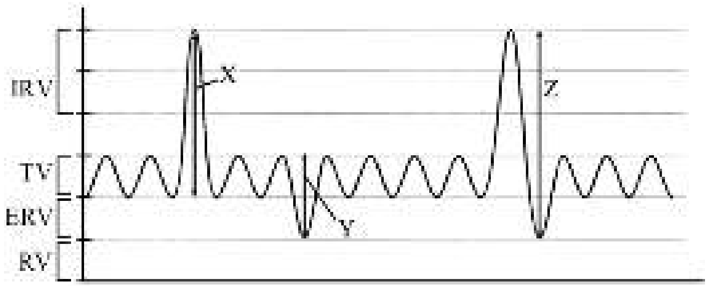
(1) A, B and C

(2) B and D

(3) Only D

(4) B, C and D

27) Given below the spirogram of a healthy person in which ‘X’, 'Y' and 'Z' represents :-



(1) IC, EC & FRC

(2) EC, IC & TLC

(3) FRC, IC & EC

(4) IC, EC & VC

28) How many of the following factors are responsible for increased P_{50} value ?
(a) Increased P_{O_2}
(b) Decreased P_{CO_2}
(c) Increased H^+ concentation
(d) Decreased pH
(e) Increased temperature

(1) One

(2) Two

(3) Three

(4) Four

29) How many of the following statement are not true?
(I) The partial pressure of oxygen in oxygenated blood is 95 mm Hg
(II) The partial pressure of oxygen in alveolar air is 104 mm Hg
(III) The partial pressure of O_2 in the alveolar air is 40 mm Hg
(IV) The partial pressure of CO_2 in deoxygenated blood is 95 mm Hg

(1) II and I

(2) I, II, and III

(3) I and II

(4) III and IV

30) Read the following statements :-
(A) High $H^{\oplus}$ concentration and higher temperature are favourable for dissociation of oxygen from oxyhaemoglobin.
(B) Partial pressure of oxygen in alveoli is higher than in tissue.
(C) The maximum volume of air a person can breath in after a forced expiration is vital capacity.
(D) In alveoli low P_{CO_2} and high P_{O_2} are for dissociation of CO_2 from carbaminohaemoglobin.
How many statements are correct ?

(1) One

(2) Two

(3) Three

(4) Four

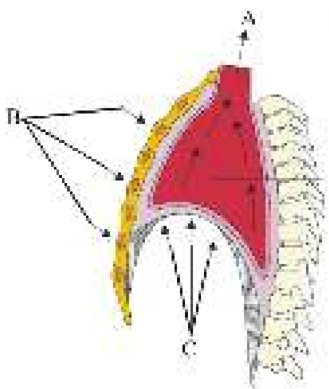
31) Under normal condition 1 litre of blood deliver ____ to tissue :-

(1) $4 m\ell O_2$

(2) $5 m\ell\ O_2$

(3) $15\ m\ell O_2$

(4) $50\ m\ell\ O_2$



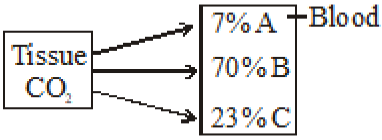
32) In the given diagram what does A, B and C depicts?

(1) A-Entry of air into the lungs
B-Ribs and sternum returned to original position C- Diaphragm contracted

(2) A-Entry of air into the lungs
B-Ribs and sternum go upwards C- Diaphragm contracted

(3) A-Expulsion of air from lungs.
B-Ribs and sternum returning to original position, C-Relaxation of diaphragm

(4) A-Air expulsion from lungs
B-Ribs and sternum go upwards C- Diaphragm relaxed

33) 

(1) A-Carboxy compound, B-Carbamino, C-Dissolved form

(2) A-Dissolved form, B-Bicarbonate, C-Carboxy compound

(3) A-Carbamino form, B-Dissolved, C-Carboxy compound

(4) A-Dissolved form, B-Bicarbonate, C-Carbamino compound

34)

Which of the following is most suitable description for trachea -

(1) Straight, extending upto the mid-thoracic cavity and divides at the level of 5^{th} thoracic vertebra.

(2) Straight, extending upto the mid-thoracic cavity and divides at the level of 9th thoracic vertebra.

(3) Curved, extending upto upper part of thoracic cavity and divides at the level of 7^{th} cervical vertebra

(4) Curved, extending upto the mid-thoracic cavity and divides at the level of 5^{th} thoracic vertebra

35) Which statement is/are correct ? (i) 97% oxygen combines with haemoglobin to form oxyhaemoglobin.
(ii) 20-25% of CO_2 combine with haemoglobin to form carboxyhaemoglobin.
(iii) Each haemoglobin can carry maximum 4 molecule of O_2.
(iv)100 ml of oxygenated blood can deliver around 5 ml O_2 to tissue under normal physiological condition.
(v) 1 dl of blood contains approx 15 gm haemoglobin.
How many statements are correct ?

(1) 2

(2) 3

(3) 4

(4) 5

36) Match the following and mark the correct options

	Animal		**Respiratory Organ**
(A)	Earthworm	(i)	Moist cuticle
(B)	Insects	(ii)	Gills

(C)	Fishes	(iii)	Lungs
(D)	Birds/Reptiles	(iv)	Tracheal tubes

(1) A-ii B-i C-iv D-iii
(2) A-i B-iv C-ii D-iii
(3) A-i B-iii C-ii D-iv
(4) A-i B-ii C-iv D-iii

37)

Which of the following statement/s is/are correct?
A- A high concentration of carbonic anhydrase is present in RBC.
B- Minute quantities of carbonic anhydrase is present in plasma.
C- Every 100 ml blood delivers approximately 4 ml of CO_2 to the tissues
D- 20-25% CO_2 is carried by haemoglobin as carbaminohaemoglobin.

(1) A, B and D
(2) A and D
(3) A, B, C and D
(4) Only A

38) What will be the P_{O_2} and P_{CO_2} in the atmospheric air compared to those in the alveolar air ?

(1) P_{O_2} lesser, P_{CO_2} higher
(2) P_{O_2} higher, P_{CO_2} lesser
(3) P_{O_2} higher, P_{CO_2} higher
(4) P_{O_2} lesser, P_{CO_2} lesser

39) In Erythroblastosis foetalis condition __A__ from the mother can leak into the blood of the foetus. This can be avoided by administering __B__ to the mother immediately after the delivery of the first child.
In the above paragraph A and B are :-

	A	**B**
1	Rh-antigen	Rh-antibody
2	Rh-antibody	Rh-antigen
3	Rh-antibody	Anti-Rh antibodies
4	Anti-Rh antibodies	Rh-antigen

(1) 1
(2) 2
(3) 3
(4) 4

40) How many cellular layers are present in the diffusion membrane in lungs ?

(1) one

(2) two

(3) three

(4) four

41) If total O_2 transported by blood from lungs to cell in a minute is 500 ml. Then the volume of O_2 transported by blood found dissolved in plasma is

(1) 970 ml

(2) 15 ml

(3) 1000 ml

(4) 30 ml

42) Some statements are given, read them and choose disorder :-
(i) Allergic condition
(ii) Difficulty in breathing
(iii) Increased I_gE
(iv) Wheezing sound

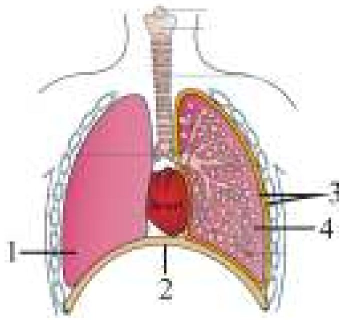
(1) Emphysema

(2) Anthracis

(3) Silicosis

(4) Asthma

43) The given figure shows the respiratory system. Identify the correct structure marked as 1,2,3 and 4 whose contraction initiated the inspiration which in turn increases the volume of thoracic



chamber in the antero–posterior axis :-

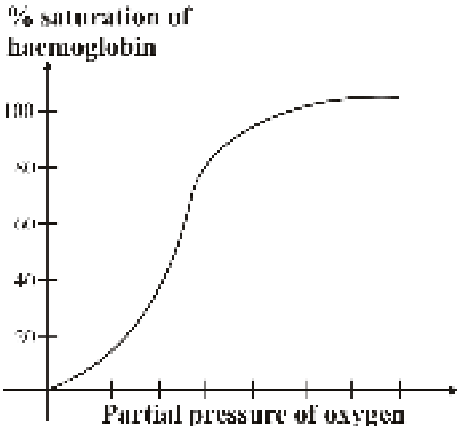
(1) 1–Lung

(2) 2–Diaphram

(3) 3–Pleural membranes

(4) 4–Alveoli

44) Refer the given figure and answer the question:-



Which of the following statement is correct regarding the above figure ?

(1) When percentages saturation of haemoglobin is plotted against the partial pressure of oxygen, a sigmoid curve is obtained

(2) Binding of oxygen with haemoglobin is primarily related to partial pressure of carbon monoxide

(3) The given graph illustrates the amount of HbO_2 as similar to Hb at different pO_2

(4) None of these

45) Match the column–I with column–II and select the correct option given below :-

	Column-I		**Column-II**
A	Tidal volume	I	2500-3000 mL
B	Inspiratory Reserve volume	II	1100-1200 mL
C	Expiratory Reserve volume	III	500-550 mL
D	Residual volume	IV	1000-1100 mL

(1) A–III, B–II, C–I, D–IV

(2) A–III, B–I, C–IV, D–II

(3) A–IV, B–III, C–II, D–I

(4) A–I, B–IV, C–II, D–III

46) The given diagram show sectional views of plasma membrane. Find out the labelled part and their function :-

	A	**B**	**C**	**D**
(1)	Tunnel protein :- polar molecule	Glycoprotein :- cell to cell recognition	Lipid :- bilayer	Cholesterol :- stability

(2)	Glycoprotein :- cell to cell recognition	Tunnel protein :- polar molecule	Lipid :- bilayer	Cholesterol :- stability
(3)	Tunnel protein :- polar molecule	Glycoprotein :- cell to cell recognition	Cholesterol :- stability	Lipid :- bilayer
(4)	Tunnel protein :- polar molecule	Lipid :- bilayer	Cholesterol :- stability	Glycoprotein :- cell to cell recognition

(1) 1
(2) 2
(3) 3
(4) 4

47) Select the incorrect statement.

(1) The cell wall determines the shape of the cell and provides a strong structural support to prevent the bacterium from bursting or collapsing
(2) The plasma membrane of prokaryotes is structurally similar to that of the eukaryotes
(3) In eukaryotic cells there is an extensive compartmentalisation of cytoplasm
(4) All eukaryotic cells are identical

48) Read the following statements :-
(A) Presence of DNA
(B) Presence of cristae
(C) Presence of ribosomes
(D) Enzymes for carbohydrate synthesis
(E) Site for oxidative phosphorylation
How many of the above statements are common for both mitochondria and chloroplast ?

(1) Five
(2) Four
(3) Two
(4) Three

49) **Assertion :** In cell membrane lipids are arranged within the membrane with the polar head towards the outer sides and hydrophobic tails towards the inner part.
Reason : This arrangement ensures that the polar tail of saturated and unsaturated hydrocarbons is protected from the aqueous environment.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

50) Which is correct option amongst the following statements?
A. Nuclear membrane, chloroplast, mitochondria, microtubules and pili are absent in prokaryotic cells.
B. Nuclear membrane, chloroplasts, mitochondria, microtubules and pili are present in eukaryotic cells.
C. Ribosomes are 70S in prokaryotic cells, chloroplast and mitochondria. They are 80s in animal cells.

(1) A and B are wrong, C is correct
(2) A is correct, B and C are wrong
(3) A and B are correct, C is wrong
(4) A and C are correct, B is wrong

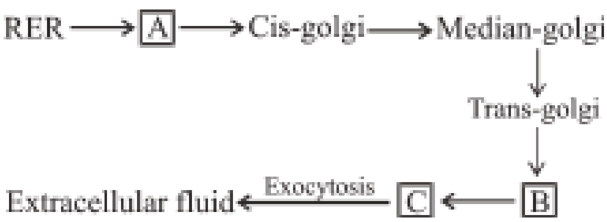
51) How many statements are **correct** ?
(A) Lipids are arranged within the membrane with the polar head towards the inner part.
(B) Membrane also contains cholesterol.
(C) Erythrocyte membrane has 52% lipid and 40% proteins.
(D) Fluid mosaic model was proposed by Singer and Nicolson.

(1) One
(2) Two
(3) Three
(4) Four

52) Choose the incorrect statement regarding plant cell.

(1) Middle lamella is chiefly made up of calcium pectate.
(2) Secondary cell wall is present outside primary cell wall.
(3) Meristematic cells have only primary cell wall.
(4) Plant cell wall mainly composed of cellulose.

53) Choose the correct for given flow chart:



	A	B	C
(1)	ER vesicle	Plasma membrane	Secretory vesicle
(2)	Secretory vesicle	Nucleus	Plasma membrane
(3)	ER vesicle	Secretory vesicle	Plasma membrane
(4)	Secretory vesicle	Plasma membrane	ER vesicle

(1) 1
(2) 2
(3) 3
(4) 4

54) From the column-I, column-II, column-III how many of the following are correctly match –

	Column-I	Column-II	Column-III
(A)	Ribosome	Composed of r-RNA and protein	Site of protein synthesis
(B)	ER	Network of tiny tubular structure scattered in the cytoplasm	Site for synthesis for lipid
(C)	Kinetochore	Disc shape structure present on secondary constriction of chromosome	Site of attachment for spindle fibres
(D)	Microbody	Membrane bound minute vesicle	Present only in plant cell

(1) 3
(2) 2
(3) 1
(4) 4

55) Which of the following is a mismatch?

(1) Endoplasmic reticulum - Divides the intracellular space into two distinct compartments.
(2) Vacuole - contains water, excretory product, sap and other materials not useful for the cell.
(3) Lysosome - Are membrane bound vesicular structures formed by the process of packaging in the golgi apparatus.
(4) Golgi apparatus - Important site of formation of lipid in plants and steroidal hormones in animal cells.

56) Match the column I with columns II & III :-

	Column I	Column II	Column III
(A)	Cytoskeleton	(i) r-RNA & proteins	(p) Fatty seeds
(B)	Peroxisomes	(ii) Only in Plants	(q) Breakdown of H_2O_2
(C)	Ribosomes	(iii) Protein filaments	(r) Protein Synthesis
(D)	Glyoxysomes	(iv) Catalase	(s) Mechanical support

Options :-
(1) A-(i)-p, B-(ii)-s, C-(iii)-q, D-(iv)-r
(2) A-(iii)-s, B-(iv)-q, C-(i)-r, D-(ii)-p
(3) A-(ii)-q, B-(iii)-p, C-(iv)-s, D-(i)-r
(4) A-(iii)-s, B-(i)-r, C-(ii)-p, D-(iv)-q

57) How many of the following will be present in a prokaryotic cell ?
Basal body, Cell Membrane, Chromatophore, Lysosome, Ribosome, Nucleolemma, Cytoplasm.

(1) Three
(2) Five
(3) Four
(4) Seven

58)

Arrange the events of cellcyle in the **correct** sequences
(i) RNA synthesis, DNA, polymerase synthesis
(ii) Replication of DNA histone protein synthesis
(iii) Tubulin protein synthesis, ATP synthesis

(iv) Karyokinesis, cytokinesis
(1) (i) → (ii) → (iii) → (iv)
(2) (ii) → (i) → (iii) → (iv)
(3) (iv) → (iii) → (ii) → (i)
(4) (iii) → (iv) → (ii) → (i)

59) In cells of higher plants, during the S-phase -

(1) DNA replication occurs in nucleus and centrioles duplicate in the cytoplasm
(2) DNA replication occurs in nucleus
(3) Centrioles duplicates in the cytoplasm
(4) Centrioles duplicates in the nucleus and DNA replication occurs in the cytoplasm.

60)

Read the following four statements (a-d)
(a) Late prophase → Golgi bodies, ER disappear
(b) Metaphase → Chromosomes arrange on equator
(c) Anaphase → Centromere split

(d) Telophase → Centriole divide
How many are correctly matched ?
(1) Three
(2) Two
(3) One
(4) Four

61) Consider the following statements (A)-(D) each with one or two blank space.
(A) __i __is marked by the initiation of condensation of chromosomal material
(B) In Oocytes of some vertebrates, ___ii___ can last for months or years
(C) During anaphase the ___iii___ divide and __iv___ start moving toward the two opposite poles.
(D) In animal cells, during s-phase __v__ replication begins in the nucleus and the ___vi___ duplicates in the cytoplasm.

Which of the following options, given the correct fills up for the respective blank number from (i) to (vi) in the statement :-

(1) (i) Prophase (ii) centromere (vi) DNA
(2) (i) Telophase (v) Centriole (vi) DNA
(3) (ii) Diplotene (iv) daughter chromosomes (vi) Centriole
(4) (i) Prophase (iv) Chromosome (v) daughter centromere

62) Identify the following stages (A) and (B) of mitosis and select the right option :-

(A) (B)

	Stage	Description	Key event
(1)	(B) Metaphase	Chromosomes highly condensed	Chromosomes move to equator
(2)	(A) Early prophase	Chromatin condensation start	Centrioles duplicates
(3)	(A) Telophase	Centromere divides	Daughter chromatids move to opposite poles
(4)	(B) Late prophase	Each Chromosome is made up of two chromatids	Golgi complex, ER, nucleolus, Nuclear envelope disappear

(1) 1
(2) 2
(3) 3
(4) 4

63) Number of chromatids in a chromosome during metaphase are :-

(1) Two chromatids in each chromosome each in mitosis and meiosis
(2) Two chromatids in each chromosome in mitosis and one chromatid in each chromosome in meiosis
(3) Two chromatids in each chromosome in mitosis and four chromatids in each chromosome in meiosis

(4) One chromatid in each chromosome in mitosis and two chromatids in each chromosome in meiosis

64) During quiescent phase of cell cycle

(1) Cell remain metabolically active
(2) Cell does not divide
(3) DNA replication can occur
(4) Both (1) and (2)

65) **Assertion :-** Crossing over is the exchange of genetic material between two homologous chromosome.
Reason :- Crossing over is an enzyme mediated process and enzyme involved is recombinase.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

66) **Assertion :** Metaphase is the stage at which morphology of chromosomes is most easily studied.
Reason : At metaphase, condensation of chromosomes is completed.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

67) **Assertion:** Homologous chromosome separate, while sister chromatids remain associated at their centromeres during Anaphase I.
Reason: Monovalent chromosomes align on the equitorial plate during metaphase I.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

68) Which one of the following statement is **incorrect**?

(1) M-phase is the most drammatic period of the cell cycle
(2) Cell division is a progressive process
(3) Prophase follows the cytokinesis
(4) Prophase is marked by the initiation of condensation of chromosomal material

69) Read the following statements and select **incorrect** statement

(1) Meiosis involves two sequential cycles of nuclear and cell division called meiosis I and meiosis II but only a single cycle of DNA replication

(2) Four haploid cells are formed at the end of meiosis II

(3) Meiosis involves pairing of Homologous chromosomes and recombination between sister chromatids of homologous cromosomes

(4) Meiosis I initiated after the parental chromosome have replicated to produce identical sister chromatids at S-phase

70) Read the following statements and find out how many statements are not correct :
(A) In telophase-I the nuclear membrane and nucleolus reappear, cytokinesis follows and this is called as dyad of cells.
(B) Meiosis increases the genetic variability in the population of organisms from one generation to the next.
(C) In zygotene stage chromosomes start pairing together and this process of association is called crossing over.
(D) Diakinesis represents transition to metaphase.

(1) Three

(2) One

(3) Two

(4) Four

71) Select incorrect statements :-
(i) Interphase represents the phase between two successive M phase.
(ii) The interphase though called the resting phase is the time during which the cell is preparing for division.
(iii) Yeast cell complete cell cycle in 90 hr.

(1) Only (i)

(2) Only (ii)

(3) Only (iii)

(4) (i) & (ii)

72) Which of the following option is correct for amount of DNA in different stages of meiosis if amount of DNA in G_1 phase is 64 pg ?

(1) Anaphase-I = 128 pg, Metaphase-I = 128 pg, Metaphase-II = 64 pg.

(2) G_2 = 128 pg, Diakinesis = 256 pg, Metaphase-I = 128 pg

(3) Pachytene = 256 pg, Metaphase-I = 256 pg, Diplotene = 256 pg

(4) S phase =128 pg, Diakinesis = 256 pg, Metaphase-I = 128 pg

73) In root tip cell there are 14 chromosomes and the DNA content is 2C, what will be the number of chromosome and amount of DNA at G_1, after S and in anaphase respectively?

	No. of chromosomes	Amount of DNA
(1)	14, 14, 14	2C, 4C, 4C
(2)	14, 28, 28	2C, 2C, 4C

(3)	14, 14, 28	2C, 4C, 4C
(4)	28, 14, 14	4C, 2C, 2C

(1) 1

(2) 2

(3) 3

(4) 4

74) Given below are two statements :
Statement-I : Cells in G_0 stage remain metabolically active but no longer proliferate unless called on to do, so depending on the requirement of the organism.
Statement-II : In animal cells, during S-phase, DNA replication begins in the cytoplasm and the centriole duplicates in the nucleus.
In the light of the above statements choose the correct answer from options given below :

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and Statement-II are incorrect.

(3) Statement-I is correct and Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

75) Cellulose do not give iodine test because :-

(1) It contain complex helices and hence degrade iodine molecules.

(2) It does not contain complex helices and hence degrade iodine molecules.

(3) It contain complex helices and hence cannot hold iodine molecules.

(4) It does not contain complex helices and hence cannot hold iodine molecules.

76) **Assertion :** Water is 70-90% of the total cellular mass.
Reason : Water is the most abundant chemical in living organism.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

77) Read the following statement (A-D) :-
(a) Each protein is a polymer of amino acids
(b) A protein is a hetero polymer and not a homo polymer
(c) Dietary proteins are the source of essential amino acid
(d) Collagen is the most abundant protein in whole of the biosphere
How many of the above statement are right :-

(1) Three

(2) One

(3) Two

(4) Four

78) Which of the following is a correct match according to secondary metabolites ?

	A-Alkaloid	B-Toxins	C-Drug
(1)	Vinblastin	Abrin	Morphine
(2)	Codeine	Ricin	Curcumin
(3)	Ricin	Morphine	Abrin
(4)	Rubber	Cellulose	Morphine

(1) 1
(2) 2
(3) 3
(4) 4

79) **Assertion :** The amino acid glycine comes under the category of non-essential amino acids.
Reason : This is due to the fact that glycine can not be synthesised in the body.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

80) Which of the following is a primary metabolite ?

(1) Anthocyanin
(2) Carotenoids
(3) Sucrose
(4) Morphine

81) Among the following neutral amino acids are?

(1) Glycine & Alanine
(2) Histidine & Arginine
(3) Glycine & Aspartic acid
(4) Glutamic acid & Tyrosine

82) Which of the following statement is correct ?

(1) Exoskeleton of arthropods have a complex heteropolysaccharide.
(2) Inulin is a polymer of fructofuranose
(3) Trypsin is a homopolymer.
(4) Lecithin is a type of glycolipid found in cell membrane

83) Read the following statement having two blanks (A) and B) :-
In oligosaccharides, monosaccharides are linked together by ....A.... . One molecule of ....B.... eliminates during bond formation. Here (A) and (B) are :-

(1) A - glycosidic bond, B - H_2S
(2) A - Posphoester bond, B - H_2O
(3) A - glycosidic bond, B - H_2O

(4) A - peptide bond, B - H_2S

84) Which of the following statement is/are **true** about enzymatic reaction ?
(A) Enzyme may change the rate of reaction
(B) Enzyme interfere with equilibrium state of reaction
(C) They lower down the activation energy of reaction
(D) Presence of enzyme is a confirmation of being a living system

(1) A & B
(2) B & C
(3) only C
(4) A,C & D

85) **Statement-I :** Enzymes generally function in a narrow range of temperature.
Statement-II : Low temperature preserves the enzyme in a temporarily inactive state.

(1) Statement I and II both are correct
(2) Statement I and II both are incorrect
(3) Only Statement I is correct
(4) Only Statement II is correct

86) Select the option which is **not correct** with respect to enzyme action :-

(1) Substrate binds with enzyme at its active site.
(2) Addition of lot of succinate does not reverse the inhibition of succinic dehydrogenase by malonate.
(3) A non-competitive inhibitor binds the enzyme at a site distinct from that which binds the substrate.
(4) Malonate is a competitive inhibitor of succinic dehydrogenase.

87)

Consider the following statements :-
(A) A number of enzymes require metal ions for their activity which forms coordination bonds with the side chains at the active sites as well as one or more coordination bonds with the substrate.
(B) The protein portion of the coenzyme is called the apoenzyme.
Select the correct option :

(1) Both (A) and (B) are true
(2) (A) is true and (B) is false
(3) Both (A) and (B) are false
(4) (A) is false and (B) is true

88) Which type of enzyme inhibition is used to control of bacterial pathogen ?

(1) Feed back inhibition
(2) Allosteric inhibition
(3) Competitive inhibition

(4) Non competitive inhibition

89) The concentration of substrate at which enzymatic activity is half of maximum, is denoted by.

(1) V_{max}

(2) $V_{max}/2$

(3) Km value

(4) ½ km

90) Match List I with List II :-

List I (Class of enzymes)		**List II (Type of reaction)**	
A.	Dehydrogenases	I	Transfer of group other than hydrogen
B.	Lyases	II	Removal of groups without hydrolysis
C.	Transferases	III	Oxidation and reduction
D.	Ligases	IV	Linking of compounds

Choose the **correct** answer from the options given below :

(1) A-II,B-III,C-IV,D-I

(2) A-III,B-I,C-IV,D-II

(3) A-II,B-IV,C-I,D-III

(4) A-III,B-II,C-I,D-IV

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ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	**1**	**3**	**2**	**4**	**4**	**4**	**3**	**4**	**3**	**3**	**2**	**1**	**3**	**2**	**3**	**2**	**1**	**3**	**3**	**3**
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	**3**	**1**	**4**	**3**	**1**	**1**	**2**	**4**	**3**	**4**	**3**	**3**	**4**	**3**	**4**	**2**	**2**	**1**	**1**	**1**
Q.	41	42	43	44	45															
A.	**1**	**3**	**1**	**1**	**1**															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	**4**	**4**	**1**	**3**	**3**	**3**	**3**	**3**	**4**	**2**	**2**	**2**	**4**	**1**	**2**	**3**	**1**	**1**	**3**	**1**
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	**4**	**1**	**3**	**4**	**3**	**4**	**1**	**2**	**3**	**2**	**1**	**4**	**2**	**3**	**3**	**1**	**1**	**3**	**3**	**1**
Q.	86	87	88	89	90															
A.	**2**	**2**	**1**	**4**	**1**															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	**4**	**2**	**1**	**1**	**3**	**3**	**2**	**2**	**1**	**2**	**3**	**4**	**3**	**2**	**4**	**2**	**4**	**2**	**4**	**1**
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	**2**	**2**	**3**	**4**	**3**	**1**	**4**	**3**	**4**	**4**	**4**	**3**	**4**	**1**	**3**	**2**	**1**	**2**	**3**	**2**
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	**2**	**4**	**2**	**1**	**2**	**2**	**4**	**3**	**3**	**1**	**2**	**2**	**3**	**2**	**4**	**2**	**2**	**1**	**2**	**1**
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	**3**	**4**	**1**	**4**	**2**	**1**	**3**	**3**	**3**	**2**	**3**	**1**	**3**	**3**	**4**	**1**	**1**	**2**	**1**	**3**
Q.	171	172	173	174	175	176	177	178	179	180										
A.	**1**	**2**	**3**	**4**	**1**	**2**	**2**	**3**	**3**	**4**										

SOLUTIONS

PHYSICS

1) **Question Explanation**
A body of mass m moves along the x-axis from rest under a constant power P. We need to determine the relationship between velocity v and time f

Concept

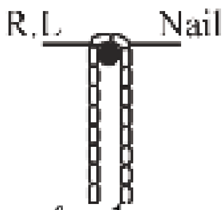
Solution:
• Power in terms of velocity and mass
P = mav   (∵ P = F.V and F =Ma)
• Acceleration is the time derivative of velocity:
$a = \frac{dv}{dt}$
• Substituting is into the power equation.
$P = mv\frac{dv}{dt}$
• Rearranging
$\frac{P}{m}dt = vdv$
• Integrating both sides
$\int vdv = \int \frac{P}{m}dt$

• Performing integration:
$\frac{v^2}{2} = \frac{P}{m}t$
• Solving for v
$v = \sqrt{\frac{2P}{m}t}$
• This shows that $\propto t^{1/2}$
Correct Answer: Option (1) $\propto\ t^{1/2}$
Question Level: Tough

2)

$W_{friction} + W_{gravity} = \Delta KE$
–mg + mgh – mgh' = 0
-μ + h = h'
–4 + 5 = h'
h' = 1 m

3) $U_i = 2\left(\frac{m}{2}\right)g\left(-\frac{\ell}{2}\right) = -mg\frac{\ell}{2}$
U_f = mg(–l) = –mgl
decrease in P.E. = increase in K.E.



$mg\frac{\ell}{2} = \frac{1}{2}mv^2$

$v = \sqrt{g\ell}$

4)

(mgsinθ + μmgcosθ) (AB)

$mg\left(\frac{h}{AB} + \mu\frac{\ell}{AB}\right) AB$

$mg(h + \mu\ell)$

5) After 10 sec,

$\frac{1}{2}mv^2 = \frac{1}{8}mv_0^2$

$\Rightarrow v = \frac{v_0}{2} = 5\,m/s$

F = ma

$-kv^2 = m\frac{dv}{dt}$

$\int_0^{10} dt = -\frac{m}{k}\int_{10}^{5}\frac{dv}{v^2}$

$[t]_0^{10} = -\frac{m}{k}\left[-\frac{1}{v}\right]_{10}^{5}$

$k = 10^{-4}$

6) $\vec{F} = -\frac{\partial U}{\partial x}\hat{i} - \frac{\partial U}{\partial y}\hat{j} = 7\hat{i} - 24\hat{j}$

$\therefore a_x = \frac{F_x}{m} = \frac{7}{5} = 1.4\,ms^{-2}$

and $\therefore a_y = \frac{F_y}{m} = -\frac{24}{5} = -4.8\,ms^{-2}$

 $v_x = a_x t = 1.4 \times 2 = 2.8\ ms^{-1}$

and $v_y = a_y t = 4.8 \times 2 = 9.6\ ms^{-1}$

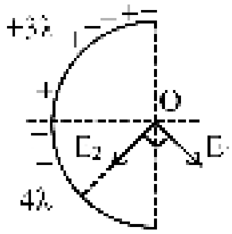
$\therefore v = \sqrt{v_x^2 + v_y^2} = 10\,ms^{-1}$

7) **1. Problem Explanation :**
Determine what the presence of a negative charge on a body indicates about the transfer of electrons or protons.

2. Concept :
An object becomes negatively charged when it gains electrons, as electrons carry a negative charge. Protons remain fixed in the nucleus and do not move during charging processes.

3. Calculation:
A negative charge results from gaining extra electrons because :
Electrons are mobile and negatively charged.
Protons cannot move between objects under normal conditions.

Final Answer :
acquired some electrons from outside



8)
$E_1 = \sqrt{2}\frac{k(3\lambda)}{R}$ & $E_2 = \sqrt{2}\frac{k(4\lambda)}{R}$

$E_0 = \sqrt{E_1^2 + E_2^2}$

$E_0 = \sqrt{2}\ \frac{k}{R}\sqrt{(3\lambda)^2 + (4\lambda)^2} = 5\sqrt{2}\ \frac{k\lambda}{R}$

9)

When body gets positive charge than its loose electron and body gets negative charge its receive electron.

10) $E = \frac{KQx}{(R^2 + x^2)^{\frac{3}{2}}}$

$x >> R$, $E = \frac{KQ}{x^2}$

11) $x = \frac{\sqrt{4Q}}{\sqrt{9Q} - \sqrt{4Q}} \times 2 = 4\text{m}$

12)

At equilibrium – $\tan\theta = \frac{F_E}{mg}$

Here, F_E same for both charges $\Rightarrow$ $\tan\theta \propto \frac{1}{m}$
if $m_1 = m_2 \Rightarrow \theta_1 = \theta_2$

13) **Question explain**
We're given diagrams showing electrostatic force directions between three charged particles. We must determine which arrangements are possible under Coulomb's law and Newton's third law.

Concept based
For any system of charges, the forces must obey Newton's third law (equal and opposite), and the net force vectors must align with the nature of attraction or repulsion between charges.
Formula used
Coulomb's law: $F = k\frac{q_1 q_2}{r^2}$
Newton's third law: $\vec{F}_{AB} = -\vec{F}_{BA}$
Calculation
Diagram I violates symmetry and balance, so it's incorrect.
Diagram II shows mutual repulsion-possible if all charges are like.
Diagram III satisfies force balance-possible.
Diagram IV violates symmetry so it is incorrect
Answer option (3)
II, III

14)

$qE = mg$

15)

$|q| \propto$ No. of ELF

16) By theory

17)

Property of metal in electrostatic conduction

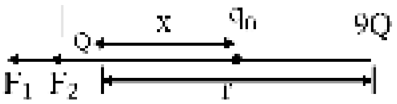
18) **Asking about:** We are given four statements about electric field lines and charge distribution on metal surfaces and asked to identify which ones are correct.
Concept: Electric field lines
Formula: No direct formula, but principles of electrostatics and boundary conditions apply;
$\vec{E} \perp$ metal surface
σ (surface charge density) varies with geometry.
Explanation:
(a) True – field lines can't penetrate a conductor.
(b) True – they are always perpendicular at electrostatic equilibrium
(c) False – not always uniform except under special symmetry
(d) True – even spherical surface can have non-uniform charge if external fields exist.
Correct option (3)

19)

$F \propto q_1 q_2$

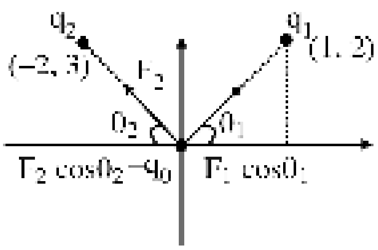
$$\frac{F'}{F} = \frac{40 \times 40}{10 \times 90} = \frac{16}{9}$$
$$F' = \frac{16F}{9}$$

20) When q_1 and q_2 are like third charge q_0 will be in equilibrium in between q_1 and q_2.

21) 

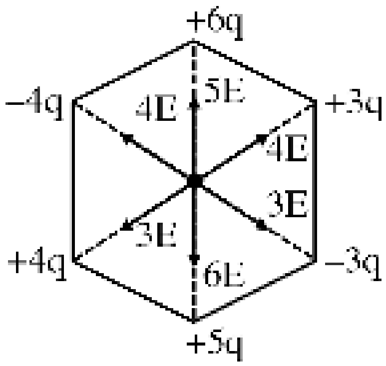
$\because F_{net} on Q = 0$
$F_1 + F_2 = 0$

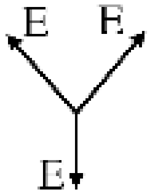
$$\frac{KQ.9Q}{r^2} + \frac{KQq_0}{(r/4)^2} = 0 \Rightarrow \left[x = \frac{\sqrt{Q}r}{\sqrt{Q} + \sqrt{9Q}} \Rightarrow \frac{r}{4}\right]$$
$$q = -\frac{9Q}{16}$$

22) 

$F_1\cos\theta_1 = F_2\cos\theta_2$

$$\frac{Kq_1q_2}{\left(\sqrt{5}\right)^2}.\frac{1}{\sqrt{5}} = \frac{Kq_2q_0}{\left(\sqrt{13}\right)^2}.\frac{2}{\sqrt{13}}$$
$$\frac{q_1}{q_2} = \frac{10}{13}\sqrt{\frac{5}{13}}$$

23) 



$E_{net} = 0$

24) Experimental analysis

25) $X^{242} \rightarrow Y^{121} + D^{121} + Q$

$Q = BE_f - BE_i$

$= 8.1\ (121 + 121) - 7.6 \times 242$

$= 242\ \{8.1 - 7.6\}$

$= 242 \times 0.5$

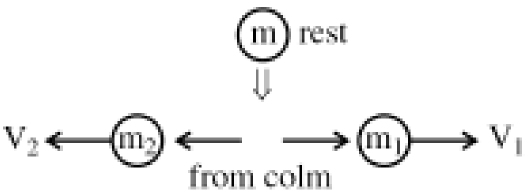
$= 121.0$ MeV

26) $\lambda = \dfrac{h}{\sqrt{3mkT}}$

$$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{T_2}{T_1}}$$

$$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{600}{300}}$$

$$\lambda_2 = \frac{1}{\sqrt{2}}\lambda_1$$



27)

$m_1V_1 = m_2V_2$

$$\frac{m_1}{m_2} = \frac{V_2}{V_1}$$

$$\frac{m_1}{m_2} = \frac{2}{3}$$

Now ratio of nuclear radii

$$\frac{R_1}{R_2} = \left(\frac{A_1}{A_2}\right)^{\frac{1}{3}}$$

$$\frac{R_1}{R_2} = \left(\frac{2}{3}\right)^{\frac{1}{3}}$$

$x = 2$

28) α β^- $2\beta^+$

$z = 92 - 2 \times 2 + 1 - 2$

$z = 87$

29) $P = nh\nu$

$$n = \frac{P}{h\nu} = \frac{2 \times 10^{-3}}{6.6 \times 10^{-34} \times 6 \times 10^{14}}$$

$n = 5 \times 10^{15}$

30) $\vec{v}_{12} = \vec{v}_1 - \vec{v}_2$

$= \tan\theta_1 - \tan\theta_2$
= constant

31) $T = \frac{2u_{rel}}{a_{rel}}$
$t = \frac{2u}{g+a}$
$\Rightarrow a = \frac{2u - gt}{t}$

32) $\vec{V}_{DC} = \vec{V}_D - \vec{V}_C = 20\hat{j}$
$\vec{V}_B - \vec{V}_C = 20\hat{i}$
$\vec{V}_B - \vec{V}_A = 20\left(-\hat{j}\right)$
Adding all
$\vec{V}_D - \vec{V}_A = 20\hat{i}$

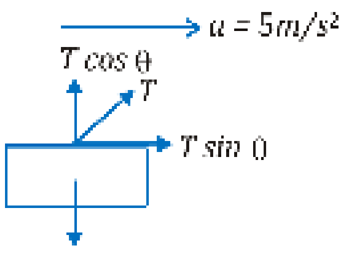
33)

$a_{rel(y)} = a_{Ay} - a_{By}$
$= g(\sin 60)^2 - g(\sin 30)^2$
$= \frac{g}{2}$
$= 4.9\ m/s^2$
vertically downwards

34)

Impulse $= \int F dt$
Area is maximum in III and IV

35) $\vec{F} = \frac{d\vec{p}}{dt}$

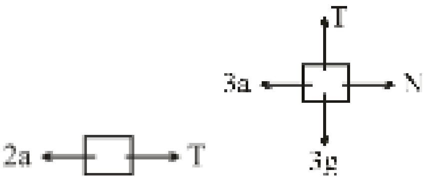


36)
$T \sin\theta = 2 \times 5$
$T \cos\theta = 20$
$\tan\theta = \frac{1}{2}$

$\cot \theta = 2$

37) Let a = acceleration

F = 15 a



T = 2a

T = 3g

2a = 3g

2a = 3g

$a = \frac{30}{2} = 15\ m/s^2$

F = 225 N

38)

(1) [Net force acting on B] → [(P) 150N]

$F_B = m_B \times a = 30 \times 5 = 150$ N

(2) [Normal reaction between A and B] → [(Q) 300 N]

$N_{AB} - 20 \times 10 = 20 \times 5$ (Calculation using block A)

$\Rightarrow N_{AB} = 100 + 200 = 300$ N

(3) [Normal reaction between B and C] → [(S) 750 N]

(Calculation using both B and A)

$N_{BC} - (20 + 30) \times 10 = (20 + 30) \times 5$

$\Rightarrow N_{BC} = (20 + 30) \times 15 = 750$ N

(4) [Normal reaction between C and elevator] → [(T) 1500 N]

(Calculation using A, B & C all three)

N_C (up) – $(20 + 30 + 50) \times 10 = (20 + 30 + 50) \times 5$

$\Rightarrow N_C$ (up) $= 100 \times 15 = 1500$ N

Aliter :

N_C (up) – N_{BC} (down) = $m_c \times a$ (up)

$\Rightarrow N_C$ (up) – 750 N = (50×5) N

$\Rightarrow N_C$ (up) – (750 + 750) N = 1500 N

39)

⇒ Acceleration relative to wedge will be along the inclined plane.

⇒ Normal reaction will be perpendicular to inclined plane.

40) (P) $(f_S)_{max} = \mu_S N = 0.6\ (2g) = 12$ N

$f_k = \mu_K N = 0.5\ (2g) = 10$ N

block won't move, a = 0, f = 10 N so 2,4

(Q) 32 > $(f_s)_{max}$ so block will slip.

[$(f_s)_{max} = 0.6\ (4g) = 24$ N

& $f_k = 0.5\ (4g) = 20$ N] **OR** (P) $F_{app} < F_L \Rightarrow a = 0$

(Q) $a = \frac{F_{app} - f_k}{m}$
(R) $F_{app} < f_L \Rightarrow a = 0$
(S) $a = \frac{F_{app} - f_k}{m}$

41)

Explain question : evaluate two statements—an **Assertion** and a **Reason**—related to the forces acting on a body in contact with a surface.
Concept : This question is based on contact force
Explanation :

$F_C = \sqrt{N^2 + f^2}$
Assertion (A) : Since the normal reaction force (N) is always non-zero for a body in contact with a surface, Therefore, contact force will always be greater than friction force
Reason (R): This is the definition of contact force. It is the vector sum (or resultant) of the normal reaction and friction force.
Final answer :- (1)

42) As forces at two ends of string are different, so tension on it decreases from 2F to F in moving from A to B.
In case of massless string, $a = \frac{2F - F}{0} = \infty$
Hence, both statements are correct.

43) $\tilde{v} = 6\hat{i} + (20 - 4t)\hat{j}$
time when $v_y = 0$
$20 - 4t = 0$
$t = 5$
$T = 2 \times t = 10$ sec
$\tan 53 = \frac{20 - 4t}{6}$
$\frac{4}{3} = \frac{20 - 4t}{6}$
$t = 3$ sec
$R = v_x \times T$
$= 6 \times 10 = 60$ m
$H = \frac{v_y^2}{2 \times a} = \frac{20 \times 20}{2 \times 4} = \frac{400}{8} = \frac{200}{4} = 50\text{m}$

44)

Total time = 2 + 6 = 8 s
$u_x = \frac{120}{6 - 2} = 30$ m/s
$R = u_xT = 8(30) = 240$ m

45) $a = \dfrac{60-(m_1+m_2+m_3)g\sin\theta\ -18}{m_1+m_2+m_3}$

$= \dfrac{12}{6} = 2\ m/s^2$

For 3 kg : $60 - 30 \sin 30° - N_1 = 3a$

For 2 kg : $N_1 - 20 \sin 30° - N_2 = 2a$

For 1 kg : $N_1 - 10 \sin 30° - F_2 = 1a$

CHEMISTRY

46)

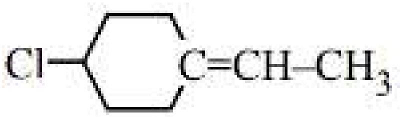
All having same R_1 S configuration

47) **Explaining the Question :-**

Identify the compound not showing geometrical isomerism

Concept:-If two some groups are present at any atom a restricted rotatory system, then geometric isomerism does not exist.

Solution :-



Here G.I. is not possible across double bonded carbons (restricted rotatory system)

Final Answer :- (4)

48)

Put option in x,y,z

49)

(3) Not matemer (Identical)

50)

H^+ attacks on more basic N atom.

51)

Theory Based question

52) Complete octate increase stability.

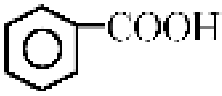
53)

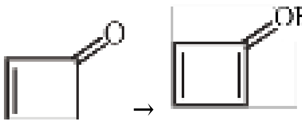
Fact : R – SO_3H > R – COOH > Ph – OH

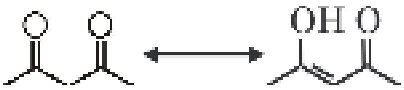
54)

depends on stability of C^+, $N^{\oplus}$, $O^{\oplus}$

55)

H–COOH >  > CH_3–COOH > H_2CO_3

56) 
(N.A.) (A.A.)

57) 
are tautomers.

58)

On the basis of C, N, O

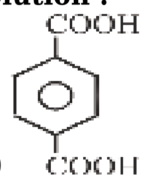
59)

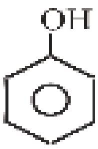
H.O.C. $\propto \frac{1}{\text{stability}}$ (When no. of carbon same.)

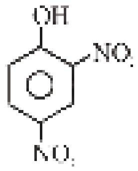
60) **Question Explanation :**
This question is asking about Which of the following not give CO_2 gas with $NaHCO_3$?

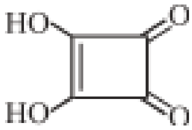
Concept :
This question is based on salt displacement reaction - Strong acid anhydr. displace whether acid salt. $NaHCO_3$ is react with these acid which are more acidic than H_2CO_3.

Solution :

(1)  * more acidic than H_2CO_3. * give CO_2 gas with $NaHCO_3$.

(2)  * less acidic than H_2CO_3. * not gives CO_2 gas with $NaHCO_3$.

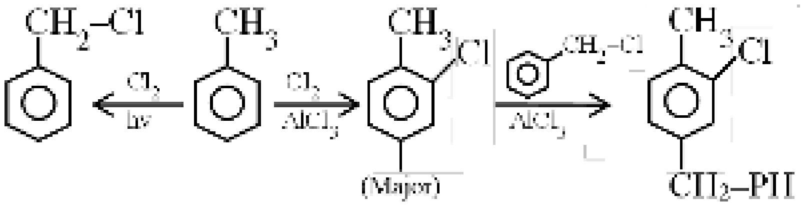
(3)  * More acidic than H_2CO_3. * give CO_2 gas with $NaHCO_3$

(4)  * More acidic than H_2CO_3. * give CO_2 gas with $NaHCO_3$.

Final Answer :
Option (2)

61)

on the basis if back bonding

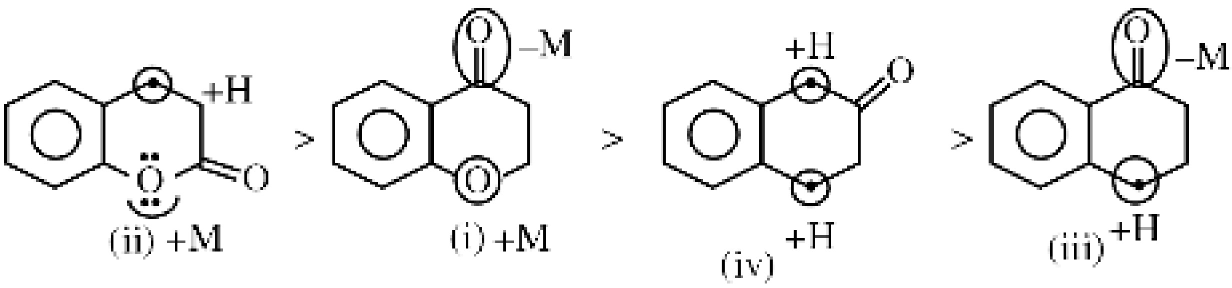
62) 

63)

Explaining the Question :- Arrange in decreasing order of reactivity towards bromination.

Concept:- Reactivity towards bromination (ESR) $\propto \frac{+M, +H, +I}{-M, -H, -I}$

Solution :- Reactivity order is :-

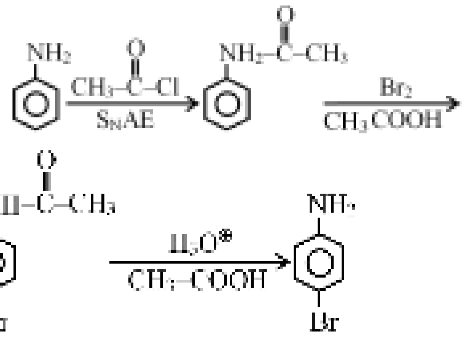


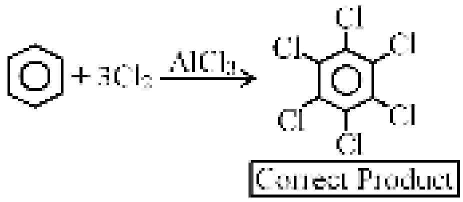
Activating groups (+M, +H) increase the rate of bromination (ESR) and deactivating groups (–M, –H) decrease the rate of bromination (ESR)

Final Answer :- Option (1)

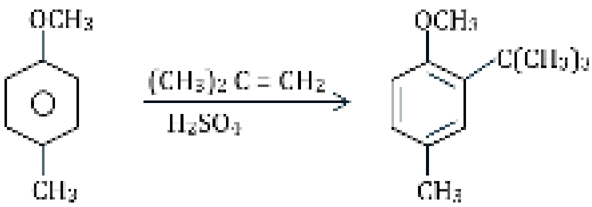
64)

On the basis of data base

65) 

66) 

67) **Asking about :** Major product of given reaction.

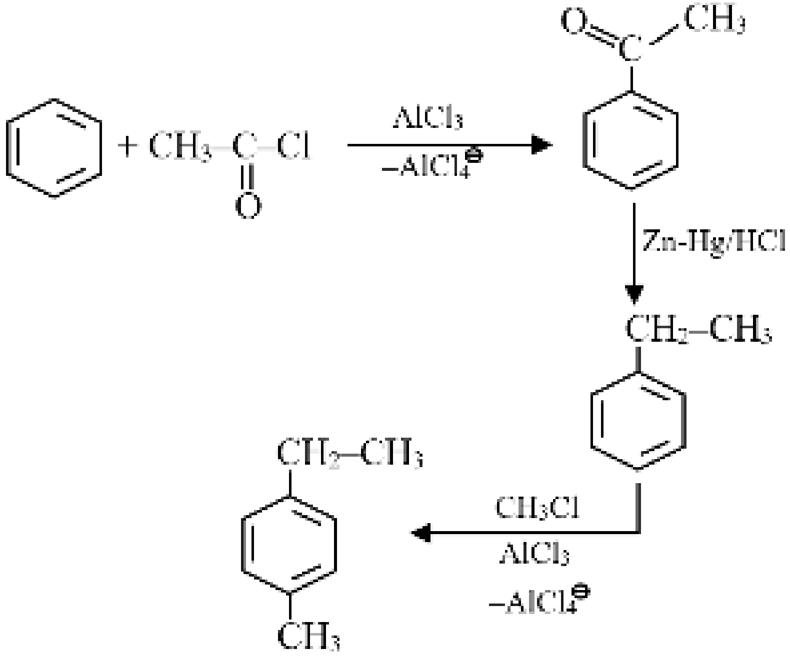


Concept :
{Hint :- –OCH_3 shows (+M, –I) effect while –CH_3 show (+H, +I) effect. +M of –OCH_3 dominates Here.}
$(CH_3)_2\,C = CH_2 + H^+ \longrightarrow (CH_3)_3 - C^{\oplus}$
Hence, the correct answer is option (1).

68) **Question Explanation :**
This question is asking about reagent in the given reaction.

Concept :
This question is based on
(i) Friedal-Craft Acylation
(ii) Clemmensen reduction reaction.
(iii) Electrophilic subsititution reaction.

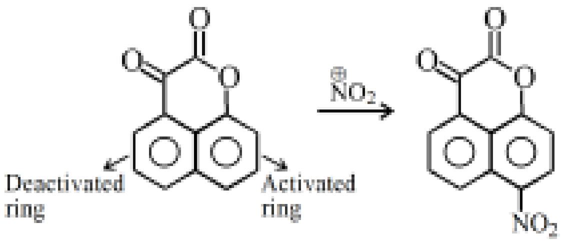
Solution :



Final Answer :
Option (3)

69)

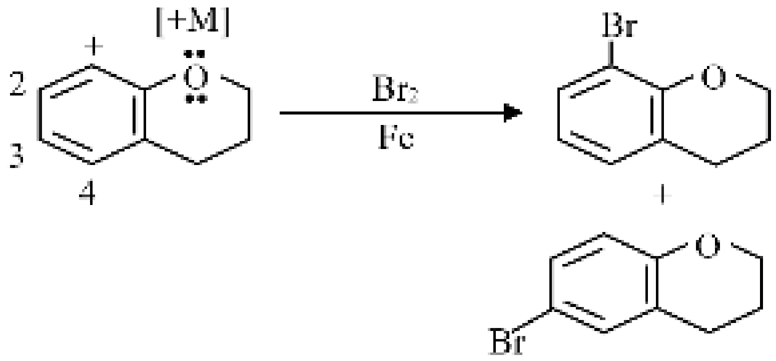
+M effect

70) 

71) **Question Explanation :**
This question is asking about position at which substitution takes place in the given compound.

Concept :
This question is based on
(1) Electrophile attack at electron rich site.
(2) +M group increase electron density at ortho-para position.

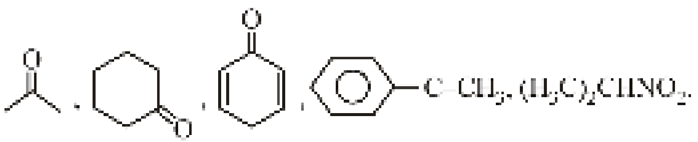
Solution :



Final Answer :
Option (4)

72)

$-OCH_3$ group has +M effect and o,p-directing group.



73) $CH_3-CH=CH-CHO$

74)

Eq (Mn_3O_4) = Eq. $FeC_2O_4.Fe(C_2O_4)_3$
mol × 2 = 1 × 9
mol = 4.5

75) **Question Explanation:** The question asks to determine the EMF (Electromotive Force) for a given concentration cell involving hydrogen gas at two different pressures, P_1 and P_2.

Concept : Nernst equation : $E_{cell} = E^0_{cell} - \frac{RT}{nF} \ln Q$
Solution :
• Anode (Oxidation): $H_2(g, P_1) \rightarrow 2H^+(aq) + 2e^-$
• Cathode (Reduction): $2H^+(aq) + 2e^- \rightarrow H_2(g, P_2)$
The overall cell reaction is: $H_2(g, P_1) \rightarrow H_2(g, P_2)$
For this reaction :
• n = 2
• $E^0_{cell} = 0$.
• $Q = \frac{P_2}{P_1}$
Substituting these values into the Nernst equation:
$$E_{cell} = 0 - \frac{RT}{2F} \ln\left(\frac{P_2}{P_1}\right)$$
$$E_{cell} = \frac{RT}{2F} \ln\left(\frac{P_1}{P_2}\right)$$

Final answer : Option 2

76)

More negative or lower is the reduction potential, more is the reducing power.

77)

Silver can not reduce Cu^{+2} ion.

78)

decrease

$$E_{cell} = E^0_{cell} - \frac{2.303RT}{nF}\log\frac{[RHS]}{[LHS]}$$

79) $E = 0.323 - \frac{0.059}{2}\log\frac{10^{-3}}{5\times10^{-3}}$

80)

$2H^+ + 2e^-\ \ H_2(g)$

$E = E^0 - \frac{0.0591}{2}\log\frac{1}{[H^+]^2}$

$E = 0 + \frac{0.0591}{2}\log[H^+]^2$

$E \Rightarrow -0.059\ pH$

$P^H_1 = 7 \quad E_1 = -0.059 \times 7 = -0.413$ V

$P^H_2 = 0 \quad E_2 = 0$

Reduction potential decrease by 0.413 V i.e. O.P $\uparrow$ es

81) $2H_2O + MnO_2 \rightarrow MnO_4^- + 4H^+ + 3e^-$ ...(III)

$Mn^{2+} + 4H_2O \rightarrow MnO_4^- + 8H^+ + 5e$ ...(II)

$MnO_2 + 4H^+ + 2e^- \rightarrow Mn^{2+} + 2H_2O$ ...(I)

Eq. (III) = I + II

$(-3F)\varepsilon^\circ = (-2F) \times 1.23 + (-5F) \times (-1.51)$

$$\varepsilon^\circ = \frac{2.46 - 7.55}{3} = \frac{-5.09}{3} = -1.69V$$

82)

Statement (A) and Statement (C) are incorrect

83) **Explanation:**

For Daniel cell, with the help of assertion and reason correct option

concept
Assertion→ In daniel cell, electrons flow from zn rod to Cu rod → true
Reason → zn rod acts as cathode and Cu rod acts as anode in Daniel cell → Incorrect
 Zn → acts as Anode
Cu → acts as cathode

Final answer = Assertion is true but the reason is false
Correct option = 3

84)

velocity of both K^+ and NO_3^- are nearly the same

85) Conductivity is conductance of all the ions present in 1 ml solution so on dilution, no. of ions per ml decreases so conductivity also decreases.

86) $(\lambda_{eq}^{\infty})BaSO_4 = (\lambda_{eq}^{\infty})BaCl_2 + (\lambda_{eq}^{\infty})H_2SO_4 - (\lambda_{eq}^{\infty})HCl$
$(\lambda_{eq}^{\infty})BaSO_4 = (x + y + z)Scm^2eq^{-1}$

87)

$Mn^{+2} + 2e^- \rightarrow Mn \qquad E° = -1.18\ V$
$2H_2O + 2e^- \rightarrow H_2 + 2OH^- \qquad E° = -0.83\ V$
Since the reduction potential of H_2O is greater than Mn^{+2} ions. So from aqueous solution of Mn^{+2} ions. So from aqueous solution of Mn salt, Mn can't be extracted.
Higher the reduction potential, greater is the ease of reduction.

88)

$$E_{cell} = E^0_{cell} - \frac{0.06}{n}\log\frac{[Al^{+3}][Ag]^3}{[Ag^+]^3[Al]}$$
$$1.50 \Rightarrow E^0_{cell} - \frac{0.06}{3}\log\frac{[Al^{+3}][1]}{[Ag^+]^3[1]}$$
$$1.50 = E^0_{cell} - \frac{0.06}{3}\log\frac{0.30}{[0.10]^3}$$
$$1.50 = E^0_{cell} - 0.02\log\frac{0.30}{0.001}$$
$$1.50 = E^0_{cell} - 0.02\log 3 \times 10^2$$
$$1.50 = E^0_{cell} - 0.02[\log 3 + \log 10^2]$$
$$1.50 = E^0_{cell} - 0.02[0.477 + 2]$$
$$1.50 = E^0_{cell} - 0.02 \times 2.477$$
$$E^0_{cell} = 1.50 + 0.04954$$
$$E^0_{cell} = 1.5496\ V$$

89) $Tl + Cu^{+2} \rightarrow Tl^{+} + Cu$
s $10^{-2}M$ $10^{-4}M$ s

$$E = E^{\circ} + \frac{0.059}{2}\log_{10}\left(\frac{Cu^{+2}}{Tl^{+}}\right)$$

90) Statement based

BIOLOGY

91)

NCERT Pg. # 197

92) NCERT Supplement

93)

NCERT Pg No. # 286,288,287,284

94)

NCERT XI Pg. # 195

95) NCERT XI Pg # 279, 280

96)

NCERT Pg. # 196

97)

NCERT Pg. # 200

98)

NCERT Pg. # 200

99)

NCERT Pg. # 200

100)

NCERT Pg. # 194

101)

NCERT Pg. # 201

102)

NCERT Pg. # 201

103)

NCERT Pg. # 197

104)

NCERT Pg. # 194

105) NCERT Pg # 279

106)

Carotid Artery supply to Brain

107)

NCERT Pg. # 200

108)

NCERT Pg. # 200

109)

NCERT-XI, Pg. # 202

110)

NCERT Pg. # 200

111)

NCERT Pg. # 200

112) **Explain Question :** Which of the following is correct representation of hepatic portal system
Concept : This question is based on portal system
Explanation : The correct pathway is Intestine → Hepatic portal vein → Liver.
Final answer : 2. Intestine → Hepatic portal vein → Liver

113)

Saline water increase B.P.

114)

NCERT Pg. # 197

115)

At high altitude RBC increases

116)

NCERT Pg. # 275

117) NCERT (XI^{th}) Pg. # 272

118)

NCERT Pg. # 189

119)

NCERT Pg. # 187

120)

NCERT Pg. # 187

121) NCERT Pg. # 274

122) NCERT-XI, Pg. # 186, Fig. 14.2

123)

NCERT XI # 274

124)

NCERT Pg. # 184

125)

NCERT Pg. # 189

126)

NCERT - XI, Page No. 268, Para - 17.1

127)

A. Question: Identify the correct statements regarding carbon dioxide transport in the blood.
B. Given Data: Four statements (A, B, C, and D) about CO_2 transport are provided.
C. Concept: This question tests knowledge of the physiological mechanisms of carbon dioxide transport in the blood.
D. Explanation:

A- A high concentration of carbonic anhydrase is present in RBC: This is correct. Red blood cells (RBCs) contain a high concentration of the enzyme carbonic anhydrase, which is crucial for the rapid conversion of CO_2 and water into bicarbonate ions (HCO_3^-) and protons (H^+). This facilitates the transport of CO_2.
B- Minute quantities of carbonic anhydrase is present in plasma: This is also correct. While the majority of carbonic anhydrase is in RBCs, small amounts are also present in plasma, contributing to CO_2 conversion there as well, though to a lesser extent.
C- Every 100 ml blood delivers approximately 4 ml of CO_2 to the tissues: This is incorrect. 100 ml of deoxygenated blood delivers approximately 4 ml of CO_2 to the lungs (not tissues). The tissues produce CO_2, which is then picked up by the blood.
D- 20-25% CO_2 is carried by haemoglobin as carbaminohaemoglobin: This is correct. About 20-25% of CO_2 binds to hemoglobin, forming carbaminohaemoglobin. This is one of the ways CO_2 is transported in the blood.
E. Final Answer: The correct answer is 1: A, B and D.

128)

A. Question: What will be the PO_2 and PCO_2 in the atmospheric air compared to those in the alveolar air?
B. Given Data: Comparison of PO_2 and PCO_2 in atmospheric and alveolar air.
C. Concept: Partial pressures of oxygen and carbon dioxide in the respiratory system.
D. Explanation:

A. PO_2 (Partial pressure of oxygen): Atmospheric air has a higher PO_2 than alveolar air. This difference is crucial for oxygen to diffuse from the alveoli into the blood. The PO_2 in the alveoli is lower because the inspired air is mixed with air already present in the lungs (which has a lower PO_2), and because oxygen is constantly diffusing from the alveoli into the blood.
B. PCO_2 (Partial pressure of carbon dioxide): Alveolar air has a higher PCO_2 than atmospheric air.

This difference is essential for carbon dioxide to diffuse from the blood into the alveoli for exhalation. CO_2 is constantly being produced by the body's cells and transported to the lungs for elimination.

E. Final Answer: 2. PO_2 higher, PCO_2 lesser

129)

NCERT Pg. # 196

130) NCERT (XI) Pg. # 273

131) NCERT (XI) Pg. # 274

132) NCERT (XI^{th}) Pg. # 275, Para-4

133)

NCERT Pg. # 184

134)

NCERT Pg. # 189

135)

NCERT Pg. # 186

136)

NCERT-XI^{th}, Pg. # 93

137) NCERT-XI^{th}, Pg. # 89, 90, 91

138)

NCERT-XI^{th}, Pg. # 96, 97

139) NCERT-XI^{th}, Pg. # 93

140)

NCERT-XI^{th}, Pg. # 89, 90, 91

141)

NCERT-XIth, Pg. # 93

142) NCERT-XIth, Pg. # 94

143)

NCERT-XIth, Pg. # 95, 96

144)

NCERT-XIth, Pg. # 95, 98, 101

145) NCERT-XIth, Pg. # 95, 96

146)

NCERT-XIth, Pg. # 102

147)

NCERT-XIth, Pg. # 89

148)

NCERT-XIth, Pg. # 126

149)

NCERT-XIth, Pg. # 121

150)

NCERT-XIth, Pg. # 122, 123

151)

NCERT-XIth, Pg. # 121, 122, 123

152) NCERT-XIth, Pg. # 123, 124

153)

NCERT-XI^{th}, Pg. # 123

154) NCERT-XI^{th}, Pg. # 121

155) NCERT-XI^{th}, Pg. # 126

156) NCERT XI Pg. # 123

157) NCERT-XI^{th}, Pg. # 126, 127

158)

NCERT-XI^{th}, Pg. # 122

159) NCERT-XI^{th}, Pg. # 126, 127, 128

160) NCERT-XI^{th}, Pg. # 126, 128

161)

NCERT-XI^{th}, Pg. # 121, 127

162) NCERT-XI^{th}, Pg. # 126, 128

163)

NCERT-XI^{th}, Pg. # 121, 123

164) NCERT-XI^{th}, Pg. # 121, 122

165) NCERT-XI^{th}, Pg. # 110

166) NCERT-XI^{th}, Pg. # 109

167) NCERT-XI^{th}, Pg. # 109, 110

168) NCERT-XI^{th}, Pg. # 108

169) NCERT-XI^{th}, Pg. # 109

170) NCERT-XIth, Pg. # 108

171) NCERT-XIth, Pg. # 108

172) NCERT-XIth, Pg. # 110

173) NCERT-XIth, Pg. # 110

174) NCERT-XIth, Pg. # 112, 113

175) NCERT-XIth, Pg. # 116

176)

NCERT-XIth, Pg. # 117

177) NCERT-XIth, Pg. # 117, 118

178) NCERT-XIth, Pg. # 117

179)

NCERT-XIth, Pg. # 116

180) NCERT-XIth, Pg. # 117, 118